

Learning Affordances of Consummatory Behaviors: Motivation-Driven Adaptive Perception

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This article introduces a formalization of the dynamics between sensorimotor interaction and homeostasis, integrated in a single architecture to learn object affordances of consummatory behaviors. We also describe the principles necessary to learn grounded knowledge in the context of an agent and its surrounding environment, which we use to investigate the constraints imposed by the agent's internal dynamics and the environment. This is tested with an embodied, situated robot, in a simulated environment, yielding results that support this formalization. Furthermore, we show that this methodology allows learned affordances to be dynamically redefined, depending on object similarity, resource availability, and the rhythms of the agent's internal physiology. For example, if a resource becomes increasingly scarce, the value assigned by the agent to its related effect increases accordingly, encouraging a more active behavioral strategy to maintain physiological stability. Experimental results also suggest that a combination of motivation-driven and affordance learning in a single architecture should simplify its overall complexity while increasing its adaptivity.

Keywords sensorimotor interaction · homeostasis · affordance · motivation · reinforcement · robotics

1 Introduction

The goal of this article is to formalize and test some of the principles required to learn object affordances, focusing on the relationship between sensorimotor interaction and the agent's internal physiological dynamics (the homeostatic loop). This work is mainly inspired by Gibson's theory of affordances (Gibson, 1986), some of whose ideas we present next. Although the notion of affordance has been controversial, the appeal of the ecological formulation to design the sen-

sorimotor loop for artificial agents has certainly not been affected (Chemero & Turvey, 2007; Cos-Aguilera, Cañamero, & Hayes, 2003a, 2004; Morineau & Frénod, 2009; Şahin, Çakmak, Doğar, Ugur, & Üçoluk, 2007). Gibson's affordances introduced the notion of directed perception, thereby suggesting the possibility of viewing an environment as a set of potentialities for action. Most of the theory's appeal for roboticists lies precisely in this *directedness* (Gibson, 1986), that is, the potentialities for action arise/disappear as elements offering the potentiality of executing a behavior from

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the animal's repertoire enter/leave its perception range. The simplicity of this formalization has been accepted by some in the autonomous robotics community, who have sought inspiration in ethological studies (McFarland, 1993; Spier & McFarland, 1996). In support of this formalization, recent neurophysiological recordings have shown that the cortical frontoparietal loops can simultaneously represent several potential options and that these compete even after a decision has been made and its related motor command has been initiated (Cisek & Kalaska, 2005; Song & Nakayama, 2007).

Therefore, a vast majority of studies on adaptation have focused on behavior planning and action selection as the main processes underlying adaptive behavior, yielding models of autonomous robots (Konidaris & Barto, 2006), cognitive science and physiology (Cisek, 2007), and reproducing different aspects and views of behavioral planning and behavior selection (Khamassi, Lachèze, Girard, & Guillot, 2006). In a complementary fashion, this article introduces a formalization of the fundamentals enabling adaptation to the environment by associating the physiological effect exerted by behavioral interactions with certain elements in the environment on the homeostatic processes, to the perception of those same elements. In other words, if the object affords to the agent the behavior currently being executed, interacting with it may exert some compensatory effect on the agent's physiology. The resulting fluctuation of the agent's internal physiological variables may then be used to gain grounded behavioral knowledge about that particular element of the environment. The continuous nature of this learning process and the ability to quantify differences and similarities between the morphology of different objects is what confers perception some of its adaptivity. Therefore, fruits that have never been seen before may, by their similarities, be recognizable as edible prior to a first interaction. Furthermore, changes in the physiological effect exerted by a particular behavior in the presence of a given element of the environment, as a result of changes in the environment (e.g., seasonal changes) or the rhythms of the animal's internal physiological processes, could be rapidly accounted for. For example, if food is no longer so readily available, the physiological effect resulting from eating may remain unchanged. In contrast, the value of this effect may increase significantly, because there is a mismatch between the internal pace of energy consumption and the previous

rate of food intake. This dynamic relates grounded knowledge from the environment to potential changes of behavior and alters the natural habits or behaviors of the individual maintaining the internal physiological variables within their viability zone (Ashby, 1965) in a manner more efficient than if perception were a fixed-input process only.

This view of affordances is inspired by perception as a process resulting from developmental and genetic adaptation, attuned by the mutuality between an animal and its surrounding environment, where the animal lives and interacts. This reflects on the elements of the environment which have become meaningful to animals of different species, always dependent on their abilities and morphology. For example, while a tree may afford us shadow and its fruits some food, it may also offer different affordances to an animal with different morphology. Furthermore, the average conditions in an environment, for example extreme drought or humidity, shape the internal physiology and therefore the general behavioral organization with which animals face the environment. The physiology of a species has developed in a given environment, the same environment where that species belongs. Altering this may severely diminish their chances of survival. Reformulating this in a more mechanistic fashion, only if the animal possesses grounded knowledge about the environment will the agent have the necessary background to manage its behaviors efficiently to satisfy its internal deficits. As formalized by Ashby, physiological stability is a necessary condition for survival (Ashby, 1965).

Additionally, the value of a potential resource depends on the generalization an animal can make of the affordances offered by similar resources. Therefore, animals use metrics of similarity between objects allowing them to extend their learned affordances from the original object they interacted and learned with to any other similar object. Furthermore, although this mechanism may not be universal across species, animals are also endowed with the additional ability to interpret the physiological effect exerted by a given behavior—often referred to as the sense of valency (Ackley & Littman, 1991). This may be explained as the ability to assess the physiological effect during and after interaction with elements from the environment, and to modulate the resulting reward. This assessment may then be used to promote or discourage the behavioral interaction with that particular object, depending

on whether the interaction is rewarding or not. Therefore, reward may be internally modulated by the animal if required (Damoulas, Cos-Aguilera, Hayes, & Taylor, 2005). For example, it may be possible to reverse the rewarding effect of eating if the animal is sated, consequently discouraging its exercise.

We have formalized these principles in an agent architecture, embedded in a situated autonomous robot, tested in a simulated environment. Since we have focused on the interaction between internal physiology and sensorimotor interaction, the agent architecture we present here encompasses three related processes: the internal *homeostatic processes* driving the animal's internal physiology; the *sensorimotor loop* relating the agent and its environment; and the *internal sense of valency* the agent is endowed with, assessing the physiological effect of the execution of a certain behavior involving certain elements of the environment. This formulation introduces the natural processes involved in affordance learning, in an abstract recreation of the process of natural, developmental adaptation to the environment.

2 Background and Related Research

The term affordance has been discussed and interpreted in several ways in the psychological literature. In its original Gibsonian conception, affordance was conceived as an evolution of the notion of valence and of the notion of *Aufforderungscharacter*—see Marrow (1969). However, unlike either of them, affordance is grounded on the principle of ecology, in the context of the interaction of an animal and its surrounding environment, thereby cutting across the phenomenological and the physical dimensions, based on the perception–action loop any animal is endowed with. Citing Gibson himself: “The object offers what it offers, because it is what it is. Duality does not exist, the physical object (the only one), possesses meaning and value to begin with” (Gibson, 1986, p. 139 (paraphrased)). Furthermore, this value is independent of the animal's internal state, for example, a piece of bread affords eating to me, whether I am hungry or not. Therefore, the Gibsonian affordance is a notion defined in the context of the interacting dynamics of an animal and its surrounding environment.

In its subsequent definitions, with the exception of Turvey (1992), who considers affordances as properties of the environment the animal interacts with,

most definitions of affordances in ecological psychology have considered an animal and its environment as the context wherein the notion of affordance is meaningful (Stoffregen, 2003). A review of the definitions of affordance, together with a novel interpretation of the same notion, can be found in Şahin et al. (2007). The main goal of this formalization is the development of an affordance-based framework for robotic architectures. Toward this end, Şahin et al. (2007) integrated previous definitions in a single formalization of affordances, composed of three perspectives: the agent's perspective, the environmental perspective, and the observer's perspective. In a way, their merit is the integration of previous definitions of affordances into a single hierarchy. The agent's perspective focuses on the affordances between an animal/agent and its surrounding environment. Because of this, it has probably been utilized by most studies using affordances in mobile robotics (Cos-Aguilera et al., 2004; Doğar, Ugur, Şahin, & Çakmak, 2008). In contrast, the observer's perspective is as a step toward establishing a common semantic of grounded knowledge, whereon symbols could be added (Harnad, 1990) and communication across several individuals may be established. Following the same principles, Nehaniv and Dautenhahn (1998) introduced the possibility of using affordances as the unit of perception to be used to facilitate imitative processes between agents of different morphology.

Despite the numerous interpretations and the formalization of Şahin et al. (2007), there are still fundamental elements of disagreement on the very essence of the notion of affordance. In particular, the use of internal representations disputes the compliance of these interpretations with Gibson's original theory (Gibson, 1986). Chemero and Turvey (2007) classified affordance interpretations into two main streams, representationalists and Gibsonians. The view of representationalists could be summarized as the necessity of having at least partial representations of objects in the environment, either representing objects from the environment or “functional representations” relating sensory patterns and behavioral responses. Conversely, a Gibsonian view suggests a direct relationship between sensory flow and behavioral responses, governed by elementary principles, often referred to as laws of control (Duchon & Warren, 1994), for example, the balance strategy or the avoid-closest used by Duchon, Warren, and Kaelbling (1995) for navigation. Although

it has been suggested that robotic architectures should try to extend from the Gibsonian perspective (Chemero & Turvey, 2007), the fact is that most elaborations of affordances have used some sort of representation. From our modest perspective, we cannot argue in favor of a single view on affordances: one of the aspects of affordances that is not understood rests precisely on the scalability from basic rules governing behavior using laws of control to navigate the environment—more typical of insects or animals in the absence of a centralized decision center—to the wide range of cognitive affordances exhibited by higher mammals. Some of the principles in the context of decision making are introduced by Konidaris and Hayes (2005).

From our perspective the notion of affordance is only meaningful in the context of an agent and its environment, viewed as two parts of a single unit, characterized by the agent's internal dynamics, the environment's natural rhythms, and their interacting dynamics. This is the context wherein we focus our research on the ecological procedure of learning object affordances. In particular, we have studied the integration between the homeostatic processes and the perception–action loop, based on the effect exerted by behavioral execution. In fact, learning affordances requires the agent to be sensitive to that particular effect, either because a behavior modifies the environment in some way or because it reflects on the agent's internal state. In other words, the effect of performing a behavior provokes a change in the environment: for example, when we hit a soccer ball, it moves opposite to the kick. At a higher description level, when we perform a consummatory behavior such as eating an apple, this has an effect on the internal level of sugar in blood, which rises as a consequence of its digestion. This same principle was first proposed by Spier and McFarland (1996) in the context of learning without cognition, which we relate to learning affordances and grounded knowledge about the environment. We argue that this method establishes the basics of the semantic relationship with the environment, integrating an agent's internal physiology with the perception–action loop in a single unit. This is further analyzed in the next section.

Most approaches are still not usable for building complex robots, indicating that affordance-based robotics may still require significant research effort to scale up to complex tasks in a satisfactory manner. None-

theless, it is fair to mention that other fields such as cognitive psychology have developed working models to learn affordances, inspired by similar principles. For example, the context created by Cooper and Glasspool (2002) explicitly built a model using affordances as attentional mechanisms to filter significant information from the environment. Again, their definition is not entirely Gibsonian because it is based on symbolic features perceived from objects, defined and engineered by the designer.

In this context the contribution of neurophysiology has significantly contributed to postulate a potential cortical representation of multiple courses of action on the frontoparietal loop in the cortex of higher mammals (Cisek & Kalaska, 2005). Furthermore, Fagg and Arbib (1998) introduced a model for the extraction of affordances from the sensory input in the context of navigation and manipulation tasks, using affordances as the link between perception and the potential action offered by every situation in the environment. More recently, Cisek (2007) introduced a model of the frontoparietal cortical loops, suggesting the simultaneous representation of several potential actions (or affordances) to be executed next.

Also, the notion of affordance has been used in related fields such as human–computer interaction (HCI; St. Amant, 1999) emphasizing the situated nature of human behavior. In this context, designers view affordances as the perceived properties of a software artifact, which indicates how it can be used (Baecker, Grudin, Buxton, & Greenberg, 1995). St. Amant (1999, p. 2) also defined them as: “a mechanism that allows or facilitates the execution of some operator. More specifically, an affordance preserves the conditions necessary for the successful completion of the operator by reducing the execution cost of other appropriate operators or by increasing the execution cost of inappropriate operators.” Together with Şahin et al. (2007), St. Amant (1999) may be one of the most complete definitions of affordance, including four separate dimensions for the concept: first, Gibsonian affordances (relationships or properties of relationships); second, our perception of several properties, the surfaces, distances, areas, textures and relationships between parts (on this fact relies the design of ecological HCI); third, the mental interpretation derived from the perceptions; and fourth, the act of performing an action itself.

3 Motivation-Based Affordances

The fundamentals of the relationship between an individual (natural or artificial) and its environment are formalized by the ecological principle (Gibson, 1986). This principle states the mutuality between the animal and the niche where it lives and evolves, because both the animal and its surrounding environment are the result of their mutual interaction and should be considered as a single entity. Therefore, later on in the text when we mention the notion of object affordance, we will refer to the behavioral relationship between the perception of an object and the potentiality of executing a behavior, only meaningful in the context of the agent and its surrounding environment, the only context wherein the affordance exists. The next paragraphs of this section describe the elements that play a significant role in learning those affordances related to consummatory behaviors: homeostasis, drives, motivation, reinforcement, reward, and physiological stability.

Homeostasis has been defined as “The condition of a system when it is able to maintain its essential variables within limits acceptable to its own structure in the face of unexpected disturbances” (Cannon, 1929, p. 210). This notion has often been referred to in the psychological literature, for example, Maturana and Varela’s (1980) process of dynamic self-regulation. In agreement with these definitions, we view homeostasis as the set of bodily self-regulation processes, which could be classified into two categories: those self-regulating the body’s internal resources independently of external stimuli, for example, the digestive process after food intake; and those relating to the external world, either by influencing the animal’s behavior or by the influence interacting with the environment on the internal physiological dynamics. Therefore, homeostasis lies at the very heart of the processes driving behavior.

Drives and Stability. Homeostasis is the context used by Hull (1943) to frame the notion of drive, capturing the status of deficit or excess of its related homeostatic variables, expressing urges for compensation, either by internal self-regulation or by external interaction. Therefore, drives are internal behavior initiators, as opposed to external stimuli. Although homeostasis has been traditionally classified as a purely internal process (Cannon, 1929), there is little doubt that both processes of internal self-regulation and external com-

pensation can be viewed as two interacting elements of a single regulatory system. In fact, an animal’s behavioral repertoire directly relates its urges for compensation to the agent’s environment, each of which requires a certain set of elements in the environment to have some effect on the internal variables, and hence complete the regulatory cycle. The capacity of regulation, external or internal, relates directly the organism’s stability and to its ability to survive in a changing environment (Ashby, 1965).

Motivation/Reinforcement. Motivation was introduced in its different flavors by McDougall (1913), Freud (1940), Tinbergen (1951), and Lorenz (1966). All of their definitions “share the idea of a substance, capable of energizing behavior, held back in a container and subsequently released in action” (Hinde, 1960, pp. 200–201). According to this, motivation intrinsically relates internal physiological drives to the behaviors, whose execution affects the intensity of their related motivations themselves. In a complementary manner, reinforcement affects the future probability of choosing a certain behavior, depending on the outcome of the interactions, influencing internal physiology. Therefore, one can view motivation (or response instigation) as acting in the direction from perception to action, and reinforcement from action to perception. Both processes modify future behavioral preferences. This mutual interaction together with the hypothesis of common neural substrate (Bindra, 1969), suggests that both motivation and reinforcement may be measured in the same unit. This interrelation between the homeostasis and the perception–action, or between motivation and reinforcement, was first made explicit by Bindra: “the effects on behavior produced by reinforcement and motivation arise from a common set of neuropsychological mechanisms, and the principle of reinforcement is a special case of the more fundamental principle of motivation” (Bindra, 1969). Bindra also suggested a physiological and anatomical substrate relating motivation, behavior, and reinforcement as processes.

Reward. Related to this, the common currency problem arises from the necessity to compare different motivational states during the process of action selection (McFarland & Sibly, 1975; Redgrave, Prescott, & Gurney, 1999). We argue that the common currency is reward. On the one hand, this would be consistent with the aforementioned notions of motivation and rein-

forcement, because reward can be viewed as the strengthening principle of the common neural substrate that reinforces or weakens the relationship between the perception of a certain object and the selected behavior. Furthermore, decision making could then be conceived as an elaboration of motivational states in terms of expected reward. On the other hand, this definition of common currency is consistent with current neurological views of hypothetical roles for the basal ganglia (Houk, Adams, & Barto, 1995), suggesting reward to be the reinforcer of behavioral tendencies. Learning can be naturally integrated in this context if neuromodulatory phenomena (Fellous, 2001, 2004; Fellous & Suri, 2003; Hebb, 1949; Usher & Daveelaar, 2002) are considered in the context of reinforcement comparison algorithms (Sutton & Barto, 1981), the reason being that these algorithms have been hypothesized to drive the learning of behavior selection in circuits in the basal ganglia (Houk et al., 1995). Hence, nigrostriatal projection and striatal cells in the basal ganglia may be the “common circuitry for motivation and reinforcement” suggested by Bindra’s hypothesis (Bindra, 1969), and could also be directly related to the process of learning affordances.

Physiological Stability. The method proposed to learn affordances should provide a stable representation between the perception of certain objects and the salience of a certain behavior, meaning its potentiality for execution. Once this grounded knowledge has been made available, behavioral responses when certain objects enter the perception range should lead to responses demonstrating a better adaptation. In the light of this reasoning, it seems sensible to use behavioral experiments to assess the quality of the learned knowledge. During a first batch of experiments (Cos-Aguilera et al., 2003a), we compared the lifespan (time until one of the internal variables escapes its viability zone) between agents selecting behaviors at random and agents selecting the most salient behavior afforded by the object nearby. The results suggested that a behavioral pattern that considers the potentialities of action afforded by the objects in the nearby environment yielded a longer lifespan. Lifespan is a behavioral measure that reflects the amount of time the agents can maintain their internal resources within their viability zone. However, it disregards additional behavioral assessments about the overall physiological stability

during the life of the agent. A temporal analysis of balance between variables should yield a more accurate description of its performance in terms of quality of life. Based on this, we have suggested the use of a physiological metric (see next section) which reflects the learning performance of the affordances of objects in their surrounding environment.

Next we describe an artificial model of internal physiology that integrates the different elements and requirements described in this section. This model is also the test bed we have used to learn affordances about consummatory behaviors in the experiments described in the following sections.

4 The Model

4.1 Overview

Next we present the model implementing the aforementioned principles that relate homeostasis and sensorimotor interaction. The model consists of three main elements (see Figure 1): an artificial physiology (top part of the figure), which is an abstraction of physiological processes of self-regulation; a modified version of the growing when required (GWR) network (bottom part of the figure), aimed at clustering the agent’s sensory input; and the affordance *learning module* (reinforcement module), that uses behavioral reinforcement to learn the potentiality of executing a behavior for a given sensory input. The running timeline of this model may be described as the sum of different concurrent processes. First, the robot interacts with the objects it encounters at random in its environment. Visual perception of the objects is used by the self-organizing feature map (GWRN) to build the world map. The closest topological node to the nearby object increases its activity. If the object is underrepresented, a new node may be added. In a concurrent fashion, as times goes by the agent’s homeostatic variables decay, thus increasing the agent’s drives. The perception–action loop is closed by executing a behavior within the agent’s repertoire, depending on the node most active in the GWR and on the internal physiological state. If the execution is successful, one or more of the agent’s internal variables may increase, and the behavioral pattern may be reinforced. Each step of these processes is described in detail in the following sections.

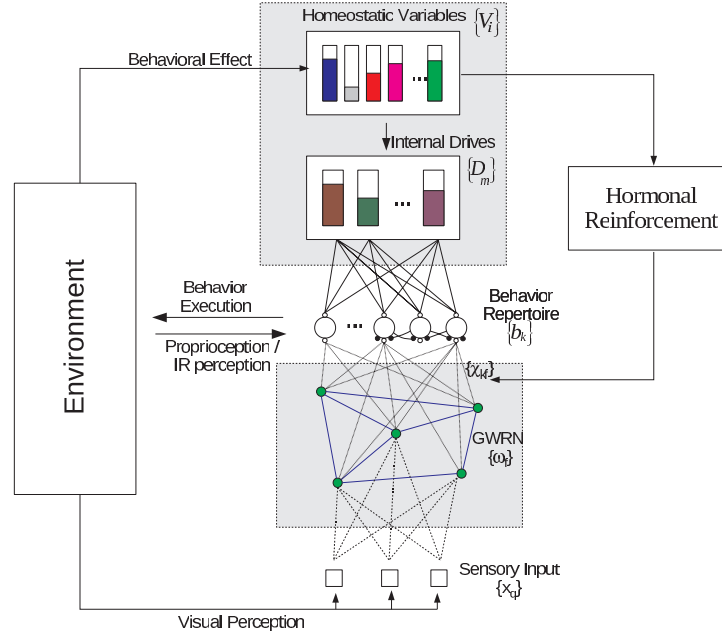


Figure 1 Illustration of the agent's architecture. The physiological variables and the internal drives are shown in the center-top of the figure, projecting directly to the agent's behaviors (center). The bottom blue network is the GWR network, which clusters visual input from the objects via the robot's camera. The *Hormonal Reinforcement* box on the right stands for the hormonal reinforcement used to learn object affordances. The circles in the center stand for the agent's behaviors, influenced by the agent's internal physiology and objects nearby, represented by the most active node of the SOFM.

4.2 Artificial Physiology

A model of artificial physiology provides the necessary context to study how behavioral feedback is integrated into the agent's internal dynamics, influencing the agent's adaptivity. The model of artificial physiology is shown in the top part of Figure 1. It consists of a set of homeostatic, survival-related variables, a set of drives that depend on the internal variables, a repertoire of behaviors, and an arbitration mechanism to resolve conflicts among competing motivations/behaviors and to choose the appropriate behavior to satisfy internal demands. The structure of this physiological model keeps a close similarity to the one used in previous studies (Cos-Aguilera et al., 2003a, 2003b, 2004).

The controlled *homeostatic variables* vary as a result of internal bodily dynamics and interactions of the agent within its environment. We model the derivative of every homeostatic variable $\dot{V}_i$, as shown by Equation 1. The first term of the equation captures the variable typical decay (at rate τ_i), and the second the increase by $\alpha(b_{ki})$ resulting from behavioral interac-

tions. $\delta(t)$ is the delta function, associated to the occurrence of a successful behavioral execution, yielding some physiological effect on the homeostatic variable.

$$\dot{V}_i = -V_i/\tau_i + \sum_k \alpha(b_{ki}) \left[\sum_{j(k)} \delta(t - t_{j(k)}) \right]. \quad (1)$$

Equation 2 shows the analytical solution to Equation 1, where $V_i(t)$ is the homeostatic variable i and τ_i the constant describing its natural decay. $\alpha(b_{ki})$ indicates the amount by which the homeostatic variable increases as a result of successful execution of behavior b_k (by interacting with an appropriate object) at times $t_j(k)$.

$$V_i(t) = V_0 e^{-t/\tau_i} + \sum_k \alpha(b_{ki}) \left[\sum_{j(k)} u(t - t_{j(k)}) e^{-(t - t_{j(k)})/\tau_i} \right]. \quad (2)$$

The homeostatic variables are abstractions representing the level of internal resources that the agent needs in order to survive. Previous studies have been constrained to the two-resource problem (Avila-García & Cañamero, 2002; Spier & McFarland, 1997), because this suffices to study some of the foundations of action selection or behavior arbitration. However, because the main topic of this article is based on the effect of behavior on the homeostatic variables, we have considered it appropriate to add an additional variable. This should introduce some desirable variability to the system while permitting an approach simple enough to study this phenomenon. We have used three basic homeostatic variables: nutrition, stamina, and restlessness. Nutrition is an abstraction for a number of elements controlling food intake (e.g., glucose level in blood). As time goes by, its level decreases, as described by Equation 1, if food/rest is not periodically consumed (the delta term indicates the moment at which the effect of successful behavior execution occurs). Stamina stands for the level of energy of the agent, which decreases over time if the agent does not regularly rest. Restlessness is an abstraction of the level of inactivity, which increases over time. Each of these variables has an optimal set point, and a range of values—their viability zone (Ashby, 1965)—for the agent to remain viable in its environment (the equivalent to remaining alive in the natural context). If the values over/underflow the upper/lower boundaries, the agent dies. Furthermore, each homeostatic variable exhibits a status of normality (when its current value approaches its ideal value or set point), deficit or excess. The homeostatic variables may express their status of deficit, or excess via a set of *drives* (Hull, 1943). Drives are abstractions denoting urges to action based on bodily needs related to self-sufficiency and survival. They monitor the divergence of each homeostatic variable with respect to its set point and initiate a process of correction. Equation 3 defines the general relationship between a drive and its related homeostatic variable(s).

$$D_m(t) = \sum_l a_{lm} (V_{op_l} - V_l(t)) + \sum_l b_{lm} \dot{V}_l. \quad (3)$$

D_m is drive m , V_l the value of the l th related homeostatic variable, and V_{op_l} the optimal (or set) value. a_{km} are the coefficients relating the variable k to the drive

m , b_{km} are the coefficients relating the rate of consumption of variable k (their derivative) to the drive m . As an initial simplification in our experiments, we have restricted this general expression to the case where each drive depends on a single homeostatic variable, furthermore making their dependence linear. To this end, the summation for the first expression will only contain a single term and the b_{km} coefficients will be zero. When a homeostatic variable diverges from its optimal point, an appropriate mechanism of compensation is triggered. In our case, the mechanism of compensation is the selection and execution of a behavior. In the general case, both the value and the rate of change of the related homeostatic variables may affect the drive, thus capturing an urgency inherently related to each internal resource. In this study, we have used three motivations: *hunger* (which controls nutrition), *fatigue* (controlling stamina), and *curiosity* (controlling restlessness). At each time step, the drives are assigned an intensity (activation level) proportional to the magnitude of the error of their controlled variable. Several drives can be active at the same time, thus establishing a competition for their related behaviors to be compensated. If in the general case every homeostatic variable may be compensated by several behaviors, in this initial simplified configuration we have related each internal variable to a particular behavior, which may be executed for compensation. For example, if food is ingested, the level of its related variable (nutrition) increases, as captured by the α_{ik} factor of Equations 1 and 2.

The overall goal for any living organism is to survive in a stable manner. Because of this, we have used Ashby's (1965) definition of *physiological balance* as the metric of performance. For the case of our agent, the physiological state is defined by the agent's internal homeostatic variables and drives, which can be described as a dynamical system to facilitate its study. Figure 2 displays an example of two-dimensional physiological space that illustrates the evolution of the agent's internal state as a result of internal resources' decays and of the execution of several behaviors. The figure also illustrates two regions within the physiological space: the viability zone, where the agent has to maintain its state to continue living; and the optimal zone, where the agent is most satisfied and its drives are minimal. In this context, the execution of a behavior can yield two different effects: either it does not modify the agent's internal state (failed execution) or

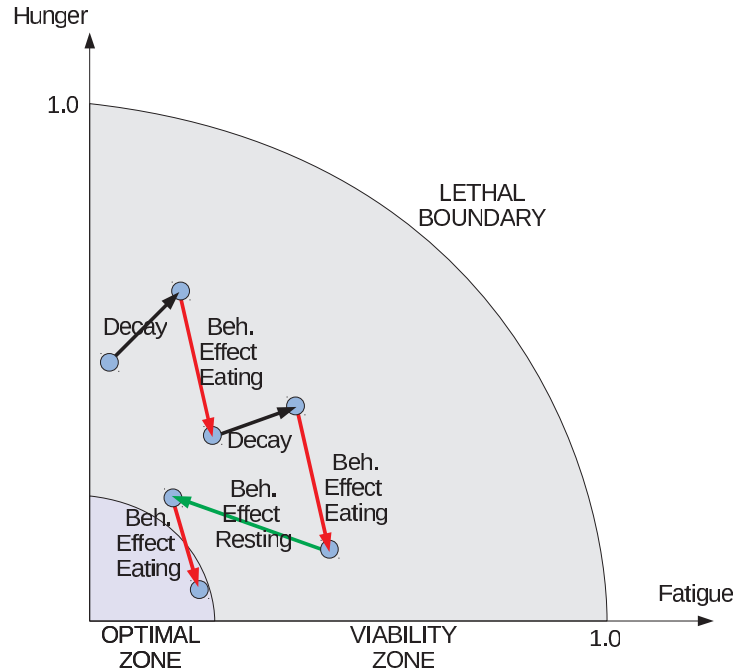


Figure 2 Depiction of a schematic representation of a 2-D physiological space. The blue dots indicate the different states of the agent's internal physiology. As time goes by, the state exhibits a decay, shifting toward the lethal boundary (black transitions). However, as consummatory behaviors are executed in the presence of an object affording that behavior, the state shifts toward the optimal zone. The optimal zone is the area where the drives are minimal. The viability zone is defined the tolerance of the drives. This is enclosed by the lethal boundary.

it diminishes the level of one or more drive (successful execution). This sort of representation is inspired by Spier's work (Spier & McFarland, 1996). The model also includes a *hormonal response*, which is internally triggered when there is a sudden variation of the internal homeostatic variables and decays exponentially afterwards. This plays a fundamental role in learning object affordances by strengthening or weakening the synapses of their neural representations (see Section 4.4).

Behaviors in this model are coarse-grained sub-systems (embedding simpler actions) that implement different competencies. The execution of a behavior modifies (increases or decreases) the levels of particular internal variables, thereby affecting the satisfaction of drives. In the general case, different behaviors can contribute to satisfy the same drive, but in our simplified model each drive can be satisfied by one behavior only—"eat" (approach edible objects) satisfies hunger, rest in a shelter satisfies fatigue, and interaction with the environment satisfies curiosity. For the behavior to have a *compensatory effect* (increment or decrement of the internal homeostatic variables), its

execution must happen in an agent–environment context where that affordance is defined.

Although this is not a study on *action selection* per se, it is aimed at assessing the working mechanisms for learning grounded knowledge about the environment in terms of physiological stability. Therefore, it is necessary to at least establish some basic rules for how the agents will resolve conflicts among competing behaviors. Two strategies have been considered to this end: *motivation driven* and *affordances × motivation* (see Figure 3 for a visual description). The former consists of selecting the behavior whose related drive exhibits the highest urge, the latter combines the averaged drives' urge and the affordances offered by the object nearby to select the next behavior. The selection mechanism works by disinhibiting the behavior whose activation is the highest while it inhibits the others. The outcome of the interaction depends on the robot interacting with an object affording the behavior selected for execution. The agent will have to learn which object offers the right affordance depending on the behavior it is trying to perform.

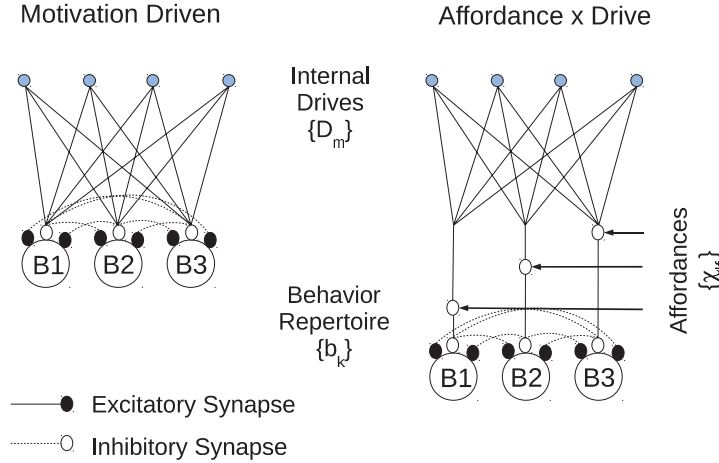


Figure 3 Strategies for behavior selection. The behavior with the highest intensity is disinhibited to be executed. Two procedures to calculate its activation are considered: motivation driven and affordance $\times$ drive.

This section has described the elements and processes integrating the agent's internal physiology (see Figure 1, center-top). The next section describes the use of an unsupervised network to cluster sensory space.

4.3 A GWRN for Sensory Perception

The very low percentage of task-relevant information in the sensory space suggests that a definition of states directly based on sensory input may not be at the right level of description to efficiently capture environment features. We have approached this problem in several related studies in a gradual manner. In a first study, we used direct feature perception, in which the agent perceived the features of objects as long as they were within range, and a two-layered architecture (Cos-Aguilera et al., 2003a). The method presented in this article is intended to be a first step to approaching the principles of learning grounded knowledge for their use in real robots. Therefore, dealing with raw sensory data and using the right level of abstraction is a requirement. Related to this, behavior-based robotics has demonstrated that the choice of behavior primitives is critical for an efficient control strategy (Konidaris & Hayes, 2005; Mataric & Brooks, 1990). Although this escapes the reach of this article, we suggest that it may also be advantageous to structure perception and behaviors as primitives that make both perception and behavior more tractable. Although this perception mechanism should ideally be based on processing sen-

sory flow, to make a test of principles we have preferred to use the more modest approach of processing a series of snapshots of nearby objects once these are in range of the robot's IR sensors.

We have designed the agent's behavior as a series of motor commands, which allow the agent to approach the object for its identification, and to execute a consummatory behavior. The snapshots, organized as 64×1 horizontal pixel images of light illumination, are normalized between 0 and 1, and fed into a variation of the growing when required network (GWRN; Marsland, Shapiro, & Nehmzow, 2002). In essence, the GWRN is an unsupervised, topological network composed of nodes and synapses, which expands or contracts depending on how well they fit sensory space. The algorithm to grow the GWRN is described next; see Marsland et al. (2002) for further detail.

- The first and the second closest nodes to the sample, nodes i and j respectively, are selected—the metric is the Euclidean distance (defined as $\sum_{i=0}^N (\vec{x}_i - \vec{y}_i)^2$).
- A synapse between i and j is grown (if not yet existing).
- The activity is calculated for each node in the network according to Equation 4.

$$a_n = \exp^{-\|\vec{\omega}_n - \vec{x}_q\|^2}, \quad n \in \{1 \dots N\} \quad (4)$$

where $\vec{x}_q$ is a sample from the sensory input and $\vec{\omega}_n$ the vector of node n . We have replaced the original linear metric by the Euclidean quadratic metric. In this manner, similar nodes in the original metric will increase their similarity, thus tending to collapse into a single node. Furthermore, with the quadratic metric, distant nodes will appear even further away. Because these nodes correspond to different objects, it should facilitate a separate representation for them. The two main parameters of the network are the activity of the node, as defined by Equation 4, and the habituation threshold (h_T), which is the maximal time allowed for a single node to find the best fitting location. If the habituation threshold is reached and the activity of the closest node is lower than the a_T threshold, it is assumed that the area neighboring that node is underrepresented, and a new node o is inserted between the best matching node r and the sample $\vec{x}_q$. The new node o is connected by two new synapses to nodes r (best matching) and g (2nd best matching). The original synapse between nodes r and g is pruned.

- If no new node is added (the activity of the node closest to the sensory input, a_r , is smaller than a_T and/or the habituation u_r is larger than the threshold h_T), the winner node r and the nodes immediately adjacent in the topology (b nodes, $b \in 1, \dots, L$) are *dragged* toward the location of the sample by an amount (ϵ_b and ϵ_r) proportional to the mismatch between them. The winner node is shifted by

$$\vec{\Delta}_r = \epsilon_b \times u_b \times (\vec{x}_q - \vec{\omega}_r) \quad (5)$$

where $\vec{\omega}_r$ is the vector representing the position of node r and $vecx_q$ is the input sample. The nodes immediately adjacent are shifted by

$$\vec{\Delta}_b = \epsilon_b \times u_b \times (\vec{x}_q - \vec{\omega}_r), \quad \forall b \in \{1, \dots, L\}, \quad (6)$$

where L is the number of adjacent nodes, $\vec{x}_q$ a sample from the sensory space, u_b and u_r two normalization constants and $\vec{\omega}_b$ the vector of node b .

- The age of all synapses connecting to the winner node is decremented by 1.
- Reduce the habituation according to

$$u_r(t) = u_0 - \frac{1}{\kappa_r} (1 - \exp(-\kappa_r t / \tau_r)) \quad (7)$$

where κ_r is a normalization constant and τ_r the decay constant for the winner node. Likewise, the counters of its L closest neighbors obey Equation 8.

$$u_b(t) = u_0 - \frac{1}{\kappa_b} (1 - \exp(-\kappa_b t / \tau_b)) \quad \forall b \in \{1, \dots, L\} \quad (8)$$

where κ_b is a normalization constant and τ_b the decay constant.

- If there are nodes with no synapses or synapses the age of which is larger than age_{max} , these are deleted.

To summarize, the parameters of the network are: activity threshold, a_T ; habituation threshold, h_T ; and the maximal synapse age, age_{max} . The age of synapses connected to frequently active nodes are reset to 0. Conversely, synapses seldom active diminish their strength by increasing their age. If the age is larger than age_{max} , the synapse is pruned. Nodes with no synapses are also pruned. Finally, the shifting coefficients ϵ_r and ϵ_r specify the dragging speed of the nodes toward the new sample.

This section has described the method applied to reduce the robot's sensory input to a tractable level of abstraction, consisting of a topological network of nodes and synapses ($\vec{\omega}_j$), which characterize the different states of the environment. In a complementary fashion, the next section introduces the structure that relates the topological nodes to the potentiality of performing a behavior and method used to learn object affordances.

4.4 Learning Affordances

Learning consists of relating the perceived sensory input to the potentiality of performing a behavior. To this end, we have grown a set of functional synapses (χ_{kj} , see Figure 4, synapses extending from GWRN topological nodes to the agent's behaviors, and Figure 1, bottom-center neural structure). Each of these synapses originates from a topological node within the GWRN, and relates to a behavior within the agent's repertoire. It is important to note that the neural structure dealing with perception contains two different sorts of syn-

the agent's internal drives and the environment, to learn grounded knowledge from the environment. As mentioned in previous sections, we have related the agent's survival to the notion of physiological stability introduced by Ashby (1965), which imposes boundaries for the agent's internal homeostatic variables for the animal to survive. The level of the agent's internal variables can be controlled by behavior execution. In agreement with these principles, we have endowed our agent with an egocentric notion of valency, implemented by the hormonal release that follows the physiological effect of executing a behavior on a homeostatic variable. This internal signal reinforces those synapses whose related behavior yields a compensatory internal effect. In other words, the role of the hormonal response is to reinforce the functional synapses between the GWRN nodes and the behaviors afforded to the agent in that context. The dynamics of the hormonal release are defined by Equation 10.

$$\begin{aligned}\dot{S} &= -S/\tau_s + \beta_i \sum_i \sum_n \delta(t - t_n), \\ t_n &= t | \dot{V}_i > X^*\end{aligned}\quad (10)$$

S is the hormonal value reinforcing the affordance synapse χ_{kf} connecting the winner node f to the behavior k just executed, V_i is the i th homeostatic variable, β_i is a normalization constant for that homeostatic variable, and t_n are the instants at which the homeostatic variable i experiences a sudden compensatory change (its derivative is larger than a large value X^*). $u(t)$ is the Heaviside step function. Since the rate of change of the homeostatic variables is several orders of magnitude smaller than that of the hormonal response ($\tau_s \gg \tau_i$), the value of the hormone will rapidly decrease to zero after the behavioral effect has been exerted. During the time S is larger than zero, the related synapse will be reinforced.

5 Experiments

The overall goal of the experiments is to test the architecture described in the previous sections to learn object affordances as a method to facilitate adaptivity by increasing the agent's internal physiological stability. To this end, the robot has been endowed with a layered behavior-based architecture, consisting of an abstract, high-level layer and a low-level action layer.

The low-level action layer controls the simpler motor commands that compose the procedural interactions to perform a consummatory behavior with a particular object. The high-level layer controls the execution of coarse-grained behaviors and is partly controlled by the agent. The wandering of the agent is driven by random navigation, switching to a consummatory behavior (eat, shelter, interact) when an object nearby is encountered. However, the agent is in control of the consummatory behavior to be executed. Once the interaction with the object has finished, the object is abandoned and the wander behavior is engaged again until another object is encountered. This set of experiments were performed with a custom designed C/C++ code, embedded in a simulated Khepera robot with a *Webots*TM simulator 4.0, K-Team (Yverdon-les-bains, Switzerland), licensed to the Institute of Perception, Action and Behavior (IPAB), University of Edinburgh.

The main focus of these experiments is to test adaptiveness of the agent's perception, which we have divided into two concurrent processes: GWRN training and learning affordances. Both processes use the same sensory input to work, consisting of images obtained during the approach to the object. The training develops a topological representation of the different angles of view of the objects of different shape and size encountered. Also, an additional neural structure, which extends its synapses off each node toward every consummatory behavior is used to learn the potentialities of performing each behavior for every object represented by that node. We refer to this overlapping neural structure as the *affordance network*, because the activation level of its synapses (χ_{kf}) indicates the potentiality of performing the related behaviors. Their level of activation is reinforced or weakened by the hormonal response following every successful object interaction. Next we describe the experimental method for both concurrent perception processes to adapt the agent's perception to the environment in a dynamic manner.

1. **Training the GWRN.** The robot behaves in a scenario where a set of objects of different shapes and sizes have been randomly distributed (see Figure 5). While no object is first detected by the robot's infrared sensors (IRs), the robot searches new resources by executing the behavior wander. When an object is detected, a consummatory behavior is executed (eat, rest, interact). Each of

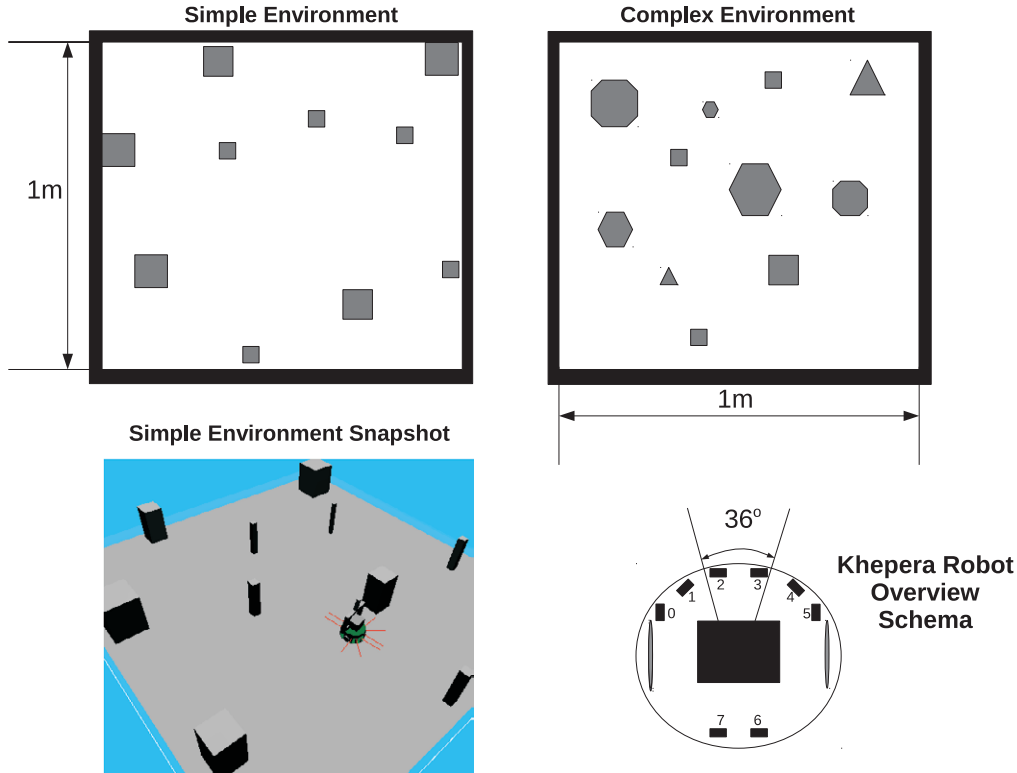


Figure 5 Schematic depiction of the simple and complex environments used for the experiments and Khepera robot overview. IR sensors are numbered from 0 to 7. The robot is also endowed with a normal vision camera (drawn as a black square) and with a gripper (not drawn) to grasp objects. Bottom-left image shows a snapshot of the actual simulation environment and of the robot in the simple environment.

these behaviors consists of a set of motor primitives, which allow the agent to execute a consummatory behavior with the nearby object. Snapshots of the object are taken near the object (within IR range), and data vectors of 64 horizontal vectors of light intensity, normalized between 0 and 1, are used to train the GWRN. Based on this data, the GWRN algorithm monitors the distance between the input data and the most similar node in the topology. If the sensory input is close enough to the node, its activity increases, otherwise a new node may be added. Two main parameters control the structure of the GWRN: the activity parameter a_j of each node and the habituation threshold h_i . The accuracy of the topology is balanced between the habituation h_i of its nodes, controlled by its decay constant τ_i and by the maximum activity a_r (see Section 4.3). Therefore, the GWRN can dynamically increase or reduce the number of

nodes, depending on the complexity of the input signal, which makes this topological network appropriate for dynamic environments.

2. **Affordance Learning.** The GWRN consists of a set of nodes (ω_j) connected by a set of synapses. Each node represents a set of objects of similar size. Therefore, in order to represent object affordances, a set of functional synapses (χ_{kj}), relate each node to every potential consummatory behavior of the agent's repertoire. We refer to these synapses as the *affordance network*. These synapses represent the affordances of the set of objects represented by each GWRN node and the behaviors the agent can perform. Next we describe the procedure to establish the activity of these functional synapses, which are initialized at a random small value.

- A series of snapshots are taken within IR range from the object. The closest GWRN node to the sensory input is identified.

- A consummatory behavior is selected at random and executed.
- The hormonal response is initiated if the effect of behavior execution reflects internally on the level of the homeostatic variables. Otherwise the hormone level remains unchanged.
- The hormonal release strengthens (or its absence weakens) the synapse of the affordance network relating the behavior just executed to the node currently active as described by a Hebbian inspired algorithm, see Equation 9.
- The simulation stops when the number of nodes is stable and a minimum of 50 interactions per node have been executed.

To measure the performance of both perception processes we have used two sets of metrics: purely statistical and behavioral/physiological. The fitting performance of the GWRN is measured for all nodes in the network, computed as defined by Equation 11.

$$\sigma_k = E\left\{\sum_{f=0}^{M-1} \sigma_{\chi_{kf}}\right\}. \quad (11)$$

where $E\{\}$ is the mean value of the summation of error variances for every functional synapse relating nodes within the GWRN to behavior k . Additionally, to assess the learning of the affordance network, we have calculated the stability of the functional synapses (χ_{kf}). Their values depend on the behaviors the objects afford to the agent and on the agent's skill, and are established by the hormonal reinforcement mechanism defined above. For example, depending on whether the orientation has been correctly matched, the object may be consumed or not. The proximity of the synaptic value to 1.0 indicates the potentiality of executing the behavior. If not, the value approaches zero.

The metrics of fitting performance and synaptic stability indicate the performance of the learning method. However, a complementary set of behavioral metrics has been designed to assess the adaptiveness of this perception method on the agent's internal milieu. In particular, we have used a viability indicator to characterize the overall physiological dynamics during the learning process and during the "life" of the agent, namely physiological stability—see Equation 9.

$$\text{Physiological Stability} = E\left\{\frac{1}{N} \sum_{m=0}^{N-1} D_m\right\} \quad (12)$$

N is the total number of drives, D_m is the m th drive, and $E\{\}$ indicates calculation of average value. This indicator is inspired by the indicators of stability introduced by Avila-García and Cañamero (2002), relating behavioral performance to the dynamics of internal physiology. The statistical and the physiological metrics provide complementary assessments about the GWRN-training and affordance-learning processes. It is important to notice that both learning processes are driven by interaction between the objects and the robot's sensors and effectors. Therefore, errors resulting from the variability of real interaction are expected to significantly influence their performance. The remainder of this section describes the results of two sets of experiments, performed in a basic and in a more complex environment, aimed at testing the performance of the proposed learning methods.

5.1 Learning Affordances in a Simple Environment

We have previously described the principles underlying affordance learning of consummatory behaviors and the learning method we have implemented. Despite its simplicity, the learning method captures the fundamentals of these principles and allows the exploration of the interactions between the different elements that compose the dynamics of interaction between the environment and the agent's internal physiology. As a first step, we tested the learning algorithm in an environment containing two sorts of objects only: smaller than the robot's gripper or larger than that. The affordances of these objects will be distributed according to their physical size (diameter), because only "thin" objects can be grasped, or shape, because only some of the objects' shapes afford the behaviors rest and interact. Within this framework, this preliminary experiment is aimed at elucidating the right parameters (node activity a_r , habituation threshold h_r , and maximal synaptic age age_{max}) to adapt to the given environment, and to test the performance of the hormonal mechanism for the synapses of the affordance network. As a rule of thumb, we have initially set the range for each parameter from the statistics of the sensory input. Thus, we have collected sensory data by

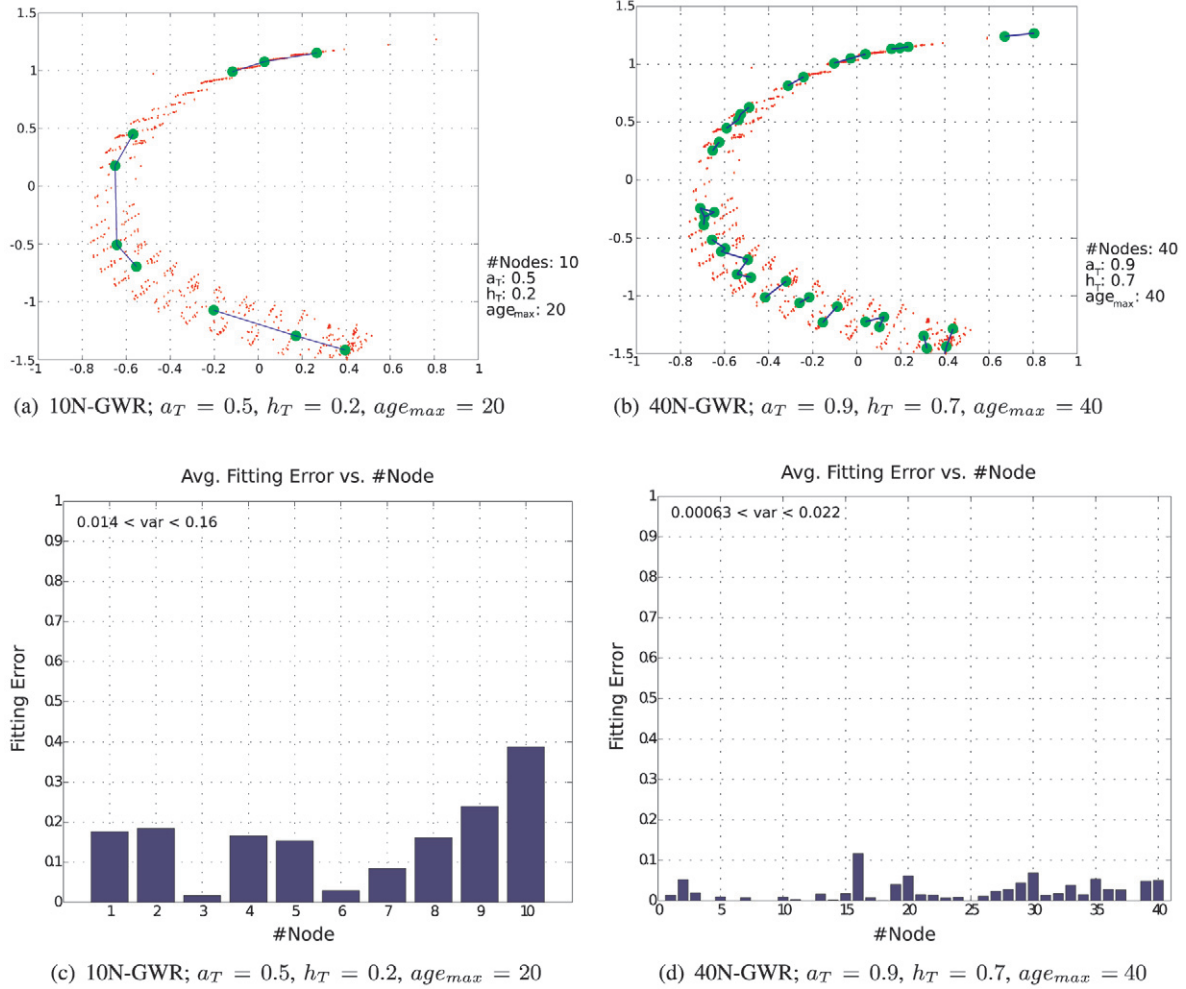


Figure 6 Depiction of two GWRN networks for the simple environment with 10 and 40 nodes, left and right, respectively. The small dots are the samples used for training. The resulting GWRN consists of a set of nodes (large circles) interconnected by synapses. The histograms underneath show the mean fitting error of the sensory space for each of node.

allowing the robot to wander in its environment for several minutes. A PCA 2-D projection of the data is shown in Figure 4, where the different views of each object appear as clusters of small dots. Using the minimal distance between clusters as rule of thumb, we have set the initial range values for the a_T threshold between 0.5 and 0.9. In a similar manner, we have used the rate of object encounter to set the h_T threshold range between 0.01 and 1.0, and of age_{max} between 5 and 40. Some of the resulting GWRN topologies are shown in Figure 6.

Figure 6a and b show 2D-PCA projections of the data (red dots) and the network topology (green dots, blue synapses). These simplified representations cap-

ture ~75% of the data variance. Figure 6c and d show the mean fitting error for each affordance synapse (χ_{kf}) for the two networks shown above, calculated according to Equation 11 for each individual node. Top left we show the ranges of variance for all nodes in the network. There are three parameters influencing the network topology: activity a_T , controlling the fitting network accuracy; habituation threshold h_T , controlling the limit of time a node is allowed to reach its final position; and the age_{max} , setting the maximum time a synapse can exist in the absence of excitation. The position of a node is final when its habituation is larger than the allowed threshold h_T . In a complementary fashion, if the activity of the node closest to the

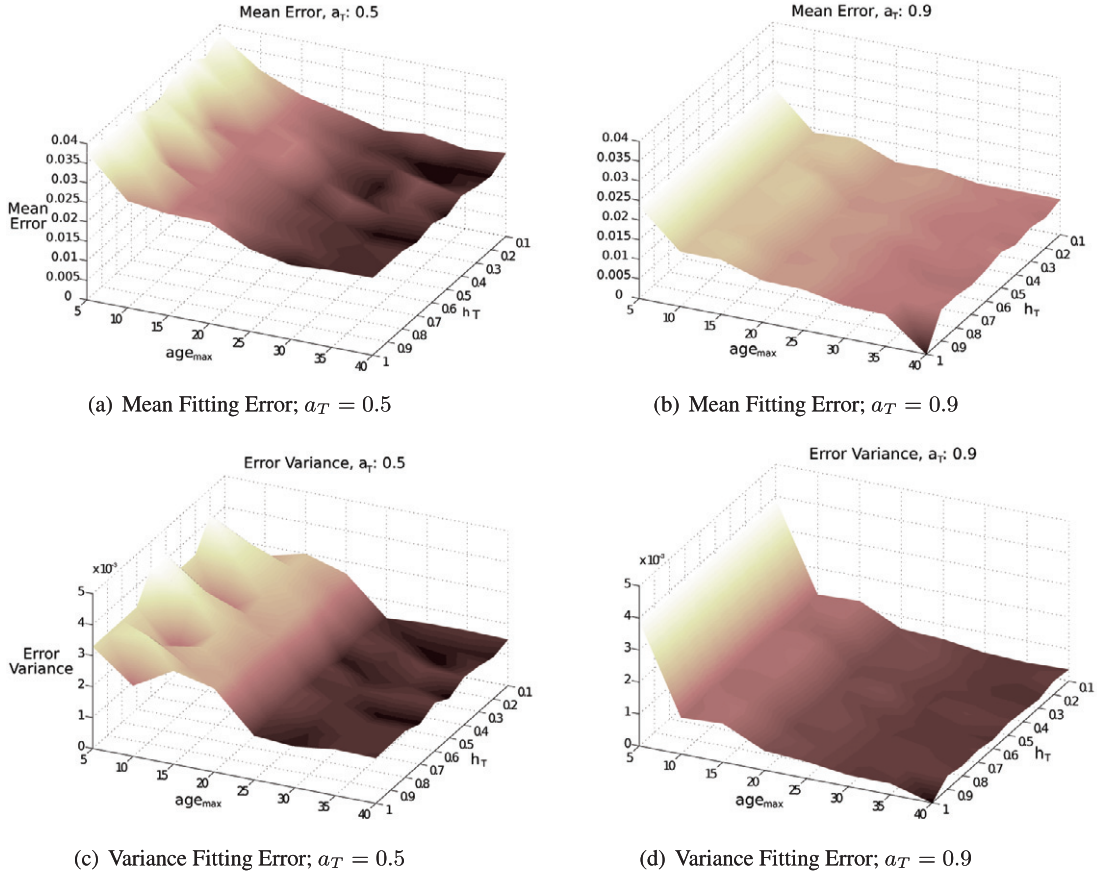


Figure 7 Fitting error for the GWRN for the case of the simple environment, mean and variance, top and bottom surfaces, respectively. The x- and y-axis are the h_T and the age_{max} parameters.

sensory input is smaller than a_T a new node may be added, because the sensory space is underrepresented. In consequence, by increasing the habituation threshold h_T we are reducing the time given for the nodes to fix their final location. Thus, if the specified accuracy has not been reached yet, some nodes will be rapidly inserted. In agreement with this, experimental results show that by increasing h_T from 0.5 to 0.7 the total number of nodes increases by five nodes. The network connectivity is mostly controlled by the age_{max} parameter. Our results show that as we increase the maximal age age_{max} from 5 to 20, the number of synapses, and therefore the accuracy of the representation, increases accordingly.

In order to assess the affordance learning method, we have calculated the mean fitting error of the GWRN during the last 25% of the simulation, averaging over 15 simulations (20,000 steps each). The results have been

represented as surfaces to cover a larger range of parameter space (see Figure 7). The x-axis is the forgetting factor h_T , the y-axis the maximal synaptic age allowed age_{max} . The z-axis shows the mean and the variance of the functional synapses. In general, the error decreased from 0.03 to 0.01, meaning a larger stability when a_T shifted from 0.5 to 0.9. For both a_T values shown, there is a plateau for synapse age_{max} larger than 20. Furthermore, the stability is not strongly dependent on the habituation threshold h_T . We have based the estimates for the learning algorithm in future simulations on the observations obtained from this preliminary experiment. In general, a_T has to be large enough to ensure sufficient accuracy (larger than 0.7), and a h_T low enough to allow a long habituation that prevents overrepresentation. Furthermore, this must be encompassed with age_{max} larger than 20 to allow this to happen before the synapses are deleted and the node

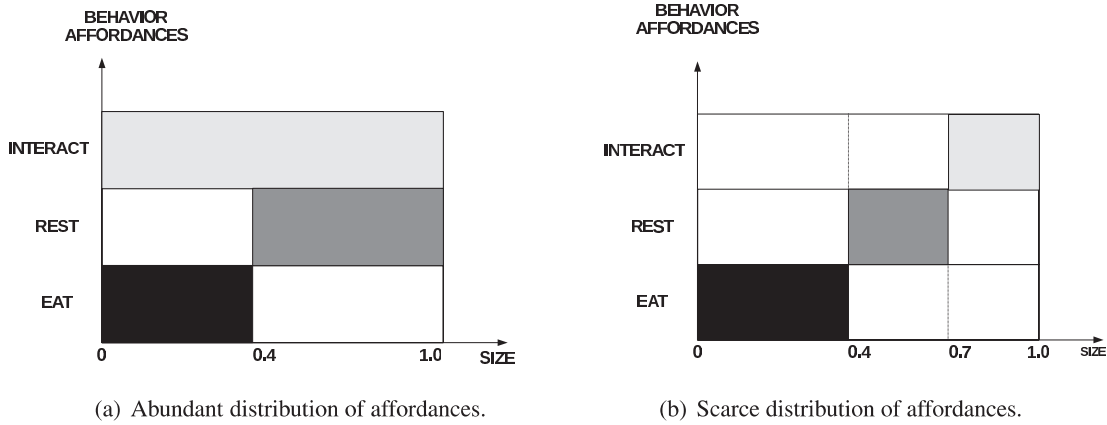


Figure 8 Abundant and scarce distributions of affordances. The affordance eat is related to the physical width of the gripper, which is 0.04. Therefore, objects smaller than this width do afford this behavior to our robot. For simplicity, we have preferred to related object size to the other behavior affordances offered to our robot. The area indicates the range of object sizes that offer that behavior. Each gray-scale indicates one behavior: eat, black; rest, dark gray; interact, light gray.

is pruned. Given these parameter constraints, we have calculated the mean fitting error versus the number of nodes. Although Figure 7 shows that there are consistent variations of the fitting error and its variance with the parameters of the network, the fact is that the average error does not significantly decrease as the number of nodes increases. This suggests that, for this simple environment, a distinction between small and large objects, obtained with a network of two nodes only, would suffice and would allow the robot to learn the affordances offered by the objects in the environment.

This may suggest that the size of the network is relatively unimportant for simple scenarios. However, the next section will demonstrate that the complexity of the network is duly related to the complexity and variability of the sensory input and the given distribution of resources of the environment. Based on this, the size and the parameters utilized for the GWRN may significantly facilitate or constrain the agent's final behavior.

5.2 Learning Affordances in a Complex Environment

The preliminary experiments in the previous scenario have demonstrated that the algorithm described previously is capable of learning grounded knowledge from the environment. This second set of experiments aims at exploring the interaction between several param-

eters relating the agent's internal physiology to the sensorimotor loop in a more complex set of environments. As mentioned in the introduction to this section, we have considered two main parameters influencing the interaction between the agent's physiology and its environment: the availability of resources in the environment and the rate of decay of the agent's internal variables. Two sorts of environments have been considered and labeled as scarce or abundant depending on the availability of affordances offered by their objects. Since the affordance eat (grasp) directly depends on the width of the object, we have extended the same relationship to the other two behaviors in a similar manner, as described in Figure 8. Every object in the *abundant* scenario will afford interaction, objects smaller than 0.04 will afford eating and objects larger than 0.04 will afford shelter. In the *scarce* scenario this is significantly restricted, because only objects whose size is between 0.04 and 0.07 will afford shelter and only objects of sizes between 0.07 and 0.1 will afford interaction. Also, an additional parameter of this experiment is the rate of decay of the agent's internal physiology, characterized as rapid and slow ($\tau = 10^{-3}/s$, $\tau = 10^{-4}/s$, respectively, for all homeostatic variables).

Also, we have proposed two strategies of behavior selection: selection of the behavior related to the drive exhibiting the highest value; and selection of the most active behavior, according to the affordance $\times$

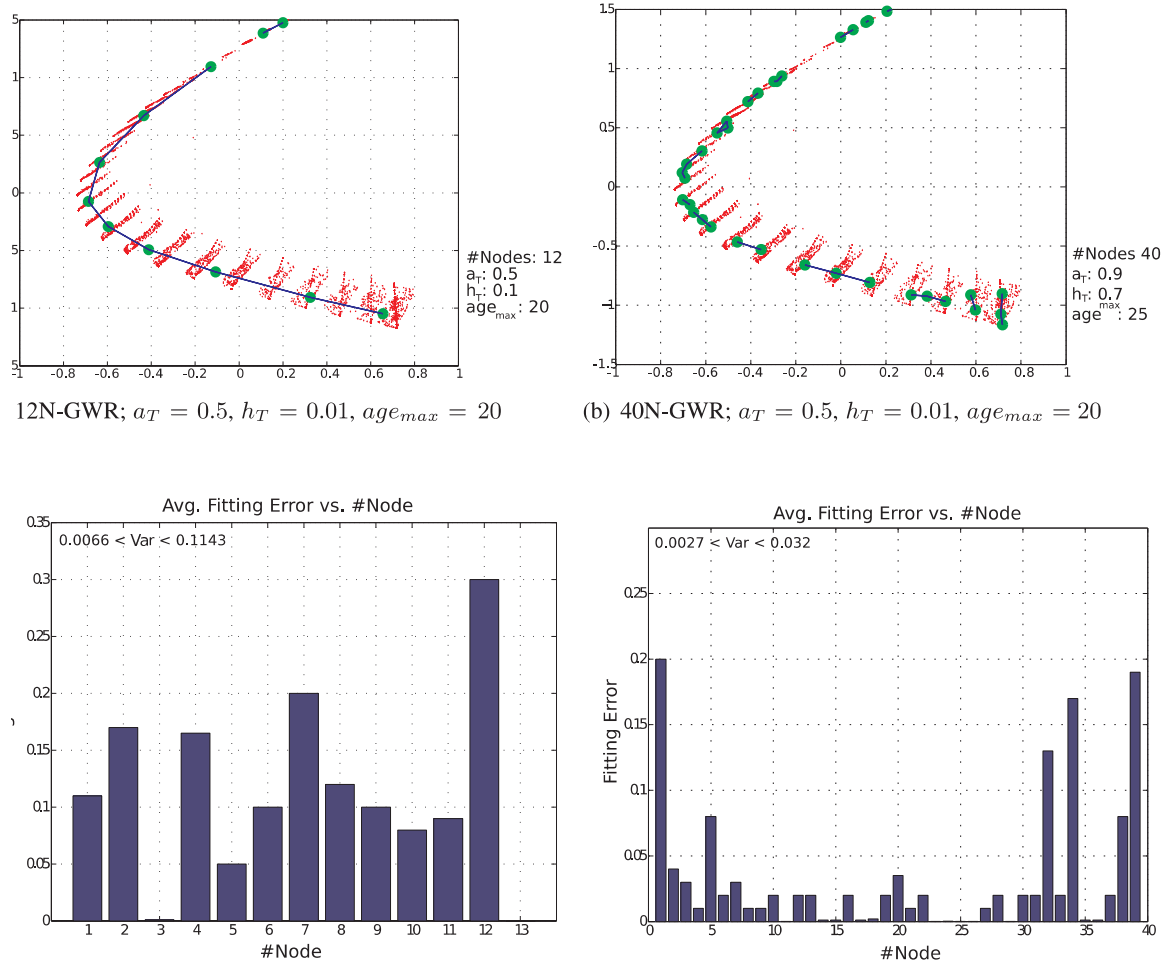


Figure 9 Top: 2-D principal component analysis (PCA) of two GWR networks built for the case of the complex environment, with 12 and 40 nodes, left and right, respectively. The small dots are the samples used to train the GWRN. The GWRN nodes are shown as circles, connected by synapses. Bottom: Histogram representing the mean fitting error for every node of the GWR network represented in top figure.

drive formula (see Section 4.2 and Figure 3 for further detail). To assess the performance of the learning mechanisms we have used two different metrics: first the fitting error of the GWRN nodes, second the overall physiological stability (see Equation 9) of the agent's internal milieu.

Preliminary Test. As for the previous set of experiments, the first goal has consisted of assessing the quality of the fitting to the input data. To this end, we have run a preliminary test consisting of ten simulations (20,000 steps each) to explore the state space of the resulting topological networks (GWRNs). At each

simulation, the agent wandered in the virtual environment and interacted with the objects nearby by selecting the behavior to execute next at random from its behavioral repertoire (see experimental method in Section 5). As for the previous set of experiments, the ranges of the parameters have been set to: a_T between 0.5 and 0.9, h_T between 0.01 and 0.1, and age_{max} between 5 and 20. Figure 9 shows some of the resulting topological structures (2D-PCA projections of data and network nodes), accompanied by the mean fitting error for each node underneath. The average error per node between both networks indicates that the number of nodes plays a significant role as the complexity of

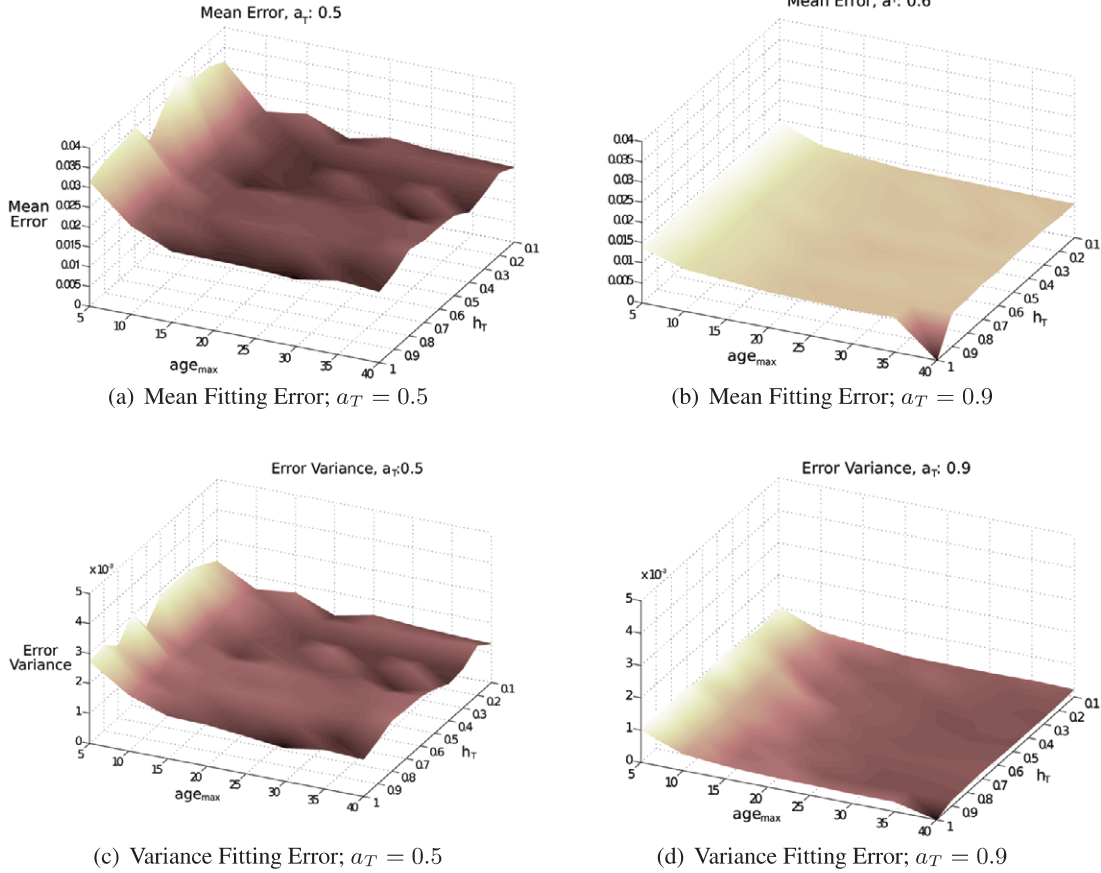


Figure 10 Mean and variance, top and bottom surface graphs respectively, of the fitting accuracy of the GWR networks. The surfaces are parametrized after the activity of the GWR and display data for $a_T = 0.5, 0.9$, left and right respectively. On x- and y-axis are plotted the age_{max} and h_T parameters respectively.

the scenario increases. We have also averaged the mean and variance of the activity of the functional synapses relating the topological nodes to the agent's behaviors (χ_{kf}). Figure 10 shows the values obtained for two networks of sizes 12 and 40 nodes, respectively. The average fitting error has decreased from 0.024 to 0.012, indicating that the larger network provides a significantly more accurate representation of the object views. In general terms, the fitting error decreases as the size of the network increases, reaching a plateau around 25 nodes (data not shown). The effects of the parameter adjustment observed are similar to those obtained from the experiments in the simple environment. As shown in Figure 10, the fitting error diminishes noticeably when the node activity rises over the threshold value a_T . Similarly, the larger age_{max} , the smaller the fitting error. As for the previ-

ous experiments, the habituation threshold h_T (y-axis) exerts little influence.

Physiological and Behavioral Assessment. To test the behavioral and physiological performance of the learning method proposed, we have assessed the physiological stability of the agent's internal milieu during the final 25% of every simulation. As mentioned in the introduction to this section, we have considered two main parameters influencing the interaction between the agent's physiology and its environment: the availability of resources, characterized by the abundant or scarce scenario, and the internal physiology, characterized by its rate of decay. Experiments exploring the parameter space for the two aforementioned parameters have been performed to measure the physiological stability of the agent's internal milieu. Also, two behavior

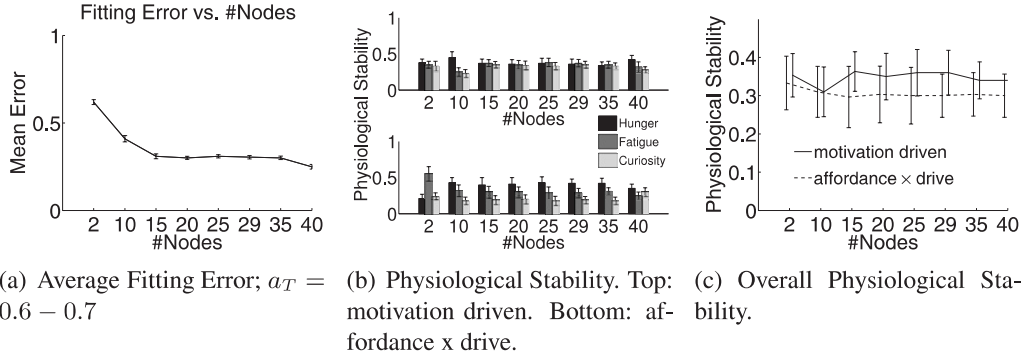


Figure 11 Statistical and behavioral metrics for agents of slow decaying physiology ($\tau = 10^{-4}$) in an environment with scarce distribution of affordances. (a) The fitting error of the GWRNs obtained for networks of size ranging from 2 to 40 nodes. (b) The physiological stability for experiments using two behavior selection strategies: motivation driven and affordance $\times$ drive, top and bottom respectively, as a function of the number of nodes. (c) The overall physiological stability for both behavior selection strategies as a function of the number of nodes. The values shown are the average over 20 simulations.

selection strategies for each set of experiments have been tested.

1. The results for the first set of experiments performed with agents endowed with a slow decaying physiology in a scarce distribution of affordances are displayed as a function of the number of nodes and of the arbitration mechanism in Figure 11. Figure 11a shows that the fitting error decreases rapidly after the GWRN expands over a few nodes. However, each individual drive (Figure 11b) and the physiological stability indicator (Figure 11c) exhibit a strong dependence on the network size and on the behavior selection strategy. In general, the drives decrease and hence the physiological stability increases as the network size expands, until a plateau is reached for networks of ~ 15 nodes or more, for the case of the affordance $\times$ drive selection strategy. As for the values obtained using the motivation-driven selection strategy, these should be considered as a performance baseline, since this strategy consists of selecting and executing the behavior associated to the drive expressing the highest urgency, disregarding the potentialities of action offered by nearby objects. Although this strategy closely monitors the agent's internal stability, it should strongly depend on the distribution of affordances. Therefore, its performance would only be expected to rival that of the multiplicative strategy when the distribution of affordances is very abundant.

2. Figure 12 shows the results for a second set of experiments in the same environment containing a scarce distribution of affordances, but with agents endowed with a fast varying internal physiology. As a consequence, the adaptation to this scenario should be driven by the urge to perform behaviors in a more efficient manner, as the rate of growth of the drives will demand compensation quicker than previously. Firstly, the fitting error in Figure 12a shows that the fitting error decreases as the number of nodes increases in two phases: first, within the range from 2–15 nodes and again from 30–40 (and possibly onwards). From the physiological metrics we have calculated, we observe that the motivation-driven strategy performs poorly, because most interactions with the objects for the environment are unsuccessful and the urges for action vary significantly faster than in the previous experiments. Therefore, both the individual drive values (Figure 12b) and the overall stability indicator (Figure 12c) are close to 1.0 (the maximal drive value), meaning that this selection strategy can hardly maintain the agent within its viability zone. Unlike this, the results obtained for the affordance $\times$ drive strategy, which combines the learned grounded knowledge from the environment potentialities of action with the internal needs of the agent, shows an increase of performance along with the size of the GWRN. In particular, as occurs with the fitting error, the overall physiological stability (Figure 12c) decreases

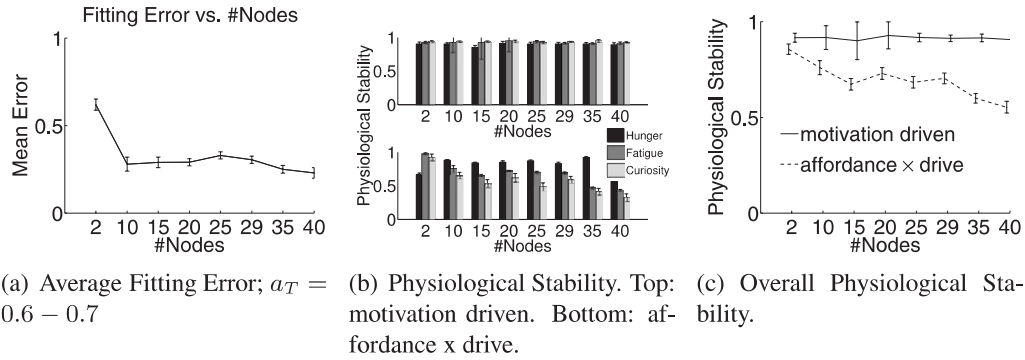


Figure 12 Statistical and behavioral metrics for agents of fast decaying physiology ($\tau = 5 \times 10^{-4}$) in an environment with scarce distribution of affordances. (a) The fitting error of the GWRNs obtained for networks of size ranging from 2 to 40 nodes. (b) The physiological stability for experiments using two behavior selection strategies: motivation driven and affordance $\times$ drive, top and bottom, respectively, as a function of the number of nodes. (c) The overall physiological stability for both behavior selection strategies as a function of the number of nodes. The values shown are the average over 20 simulations.

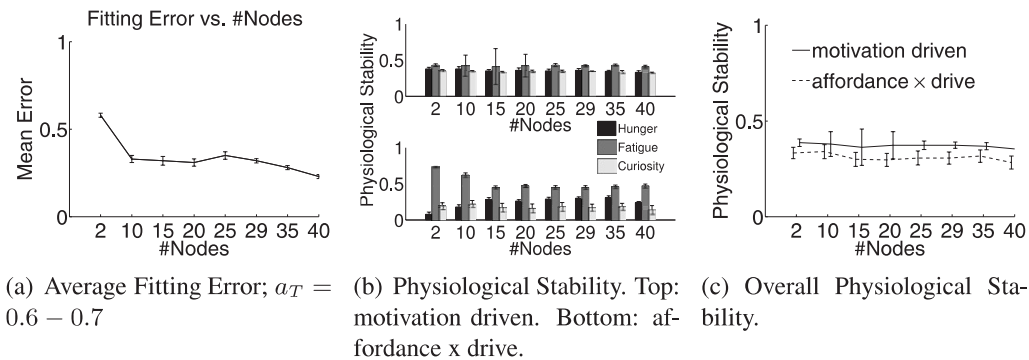


Figure 13 Statistical and behavioral metrics for agents of fast decaying physiology ($\tau = 10^{-3}$) in an environment with abundant distribution of affordances. (a) The fitting error of the GWRNs obtained for networks of size ranging from 2 to 40 nodes. (b) The physiological stability for experiments using two behavior selection strategies: motivation driven and affordance $\times$ drive, top and bottom, respectively, as a function of the number of nodes. (c) The overall physiological stability for both behavior selection strategies as a function of the number of nodes. The values shown are the average over 20 simulations.

first between 2 and 15 nodes and again as the GWRN size increases from 29 to 40 nodes.

- Finally, we have also performed a set of experiments for the abundant environment with agents, both with a slow and fast decaying physiology. We only show the results from the latter case in Figure 13. As for the two former cases, Figure 13a indicates that the GWRN fits the objects in the environment in two stages, as the number of nodes increases from 2 to 15, and then from around 29 to 40. Because the availability of resources in this

environment is larger than for the scarce case, the requirement of efficiency will decrease accordingly. Therefore, the motivation-driven strategy can already provide a stable physiology in this case (see Figure 13b top and continuous line in Figure 13c), although the rate of successful interaction is worse than for the other selection strategy. The overall performance for agents endowed with the affordance $\times$ drive behavior selection strategy indicates that using grounded knowledge is still advantageous in this abundant environment.

Therefore, these results demonstrate that the availability of resources in the environment and the rate of decay of the agent's homeostatic variables strongly depend on one another, and may dramatically influence the agent's chances of survival. As the variables' rate of decay increases, the demand for effective and frequent behavioral interactions will gradually become more urgent. This is reinforced by the results shown in Figure 12, which result from the case of a scarce distribution with fast decaying variables. These highlight that the physiological stability strongly depends on the rhythms of the agent's internal physiology and that having a mechanism of adaptive perception, integrating perception, physiology, and action in a single dynamics provides an advantageous mechanism for adapting to the environment. These results also suggest that the mechanisms proposed can be used to learn the affordances of consummatory behaviors offered by objects in the environment. In other words, to learn grounded knowledge from the objects in the environment.

5.3 Theoretical Formalization

As a whole, the results of the previous section demonstrate that there is a close relationship between the dynamics of the agent's internal physiology and the resources of the environment. This final section aims at providing an analytical formulation of this interdependence. To this end, we have formalized the dynamics between the agent and its environment by formulating the lethal boundary as a function of the decay rate of the agent's homeostatic variables and the distribution of resources of the environment in a probabilistic manner. As suggested by Ashby's principle of physiological stability, the survival of an agent will depend on its ability to maintain every homeostatic variable within its viability zone. If any of these variables decays under its critical value, the agent will die. Therefore, for every agent there will be a variable that is most critical, either because of its fast decaying rate or because of the difficulty of finding objects in the environment that afford the behavior contributing to its compensation. For that particular variable i , we can derive the critical time it takes to reach the lethal boundary. Elaborating from Equation 2, the critical time t_{max} equals the expression shown by Equation 13.

$$t_{max} = -\tau_i \log \left(\frac{V_i^*}{V_i^* + b_{ki}} \right) \quad (13)$$

Where t_{max} is the time variable i takes to reach its lethal boundary V_i^* in the absence of successful behavior interactions. b_{ki} is the effect exerted by a successful behavior interaction and τ_i the homeostatic variable decay rate. Therefore, for the agent to survive, the interactions with an object that affords its compensatory behavior must occur more often than t_{max} .

If we assume that the availability of resources of an environment is uniform, it is possible to model the statistics of successful interactions as a Poisson process with parameter λ_i . If we also assume that the interactions are always successful, the average time to compensate the variable i will equal $1/\lambda_i$. Based on these two assumptions, we may derive an analytical estimate of the lethal boundary for that particular agent. In other words, the probability of survival, as described by Equation 14, equals the probability that the time between compensatory interactions T were smaller than t_{max} , in other words, the probability that the number of compensatory interactions within a t_{max} interval were larger than 0.

$$\begin{aligned} P_{Survival} &= P\{T < t_{max}\} = P\{n > 0\} \\ &= 1 - P\{n = 0\} = 1 - e^{-\lambda_i t_{max}} \end{aligned} \quad (14)$$

Where T is the interval between interactions, and n the number of interactions during that interval. If we fix the probability that at least a compensatory interaction occurs to an arbitrary value p (probability of survival), it becomes straightforward to derive an analytical boundary for the distribution of resources the environment has to offer (characterized by the λ_i parameter of the Poisson process) for the agent to survive (see Equation 15).

$$\begin{aligned} \lambda_i &= -\frac{1}{t_{max}} \log(1 - p) = \frac{\log(1 - p)}{\tau_i \log \left(\frac{V_i^*}{V_i^* + b_{ki}} \right)} \\ &= \log \left(\frac{V_i^*}{V_i^* + b_{ki}} \right)^{\tau_i (1 - p)} \end{aligned} \quad (15)$$

Therefore, the agent will survive in environments in which the average number of interactions is smaller than $1/\lambda_i$. In this light, the distributions used in the previous experiments, scarce and abundant, may be related, respectively, to large and small λ_i values. Certainly, this simple parametrization does not include an

explicit characterization of the abilities of the agent, such as its moving speed, the agent's behavior selection strategy or the reliability of its interactions. However, it does illustrate the limitations of the relationship between the environment and the agent in a quantitative manner, and therefore their interdependence.

6 Discussion

Although adaptation is probably the most studied topic in the adaptive behavior community, most attention has been devoted to the study and design of increasingly complex decision-making algorithms. Certainly, the lifespan and the quality of life of an agent will depend on its ability to plan and decide appropriately. However, in order to do that, any living being needs to also know how to prioritize responses to compensate its internal needs. To this end, it also has to know which elements to interact with to obtain the desired effect. This is the fundamental relationship between the agent and its surrounding environment that we have formulated as a process that dynamically learns grounded knowledge about the environment. In other words, this article has proposed a formulation of the agent–environment dynamics to learn affordances of consummatory behaviors. The learning is driven by a hormonal process controlled by fluctuations in the agent's internal physiology, which result from interactions with objects in the surrounding environment. A similar process had been previously introduced in the *drk* model (Spier & McFarland, 1996, 1997) to explain the devaluation effect (Kriekhaus & Wolf, 1968) in the absence of cognition by learning the compensatory effect of several stimuli (*d-values* in their notation) using a variation of the delta rule (Widrow & Hoff, 1960). However, our architecture introduces several fundamentally different elements. First, the control of the learning process is indirectly controlled by the hormonal release. The reason for us separating the homeostatic processes and the actual learning process is that the physiological effect resulting from behavior execution requires some additional assessment. For example, while the effect of consuming food is beneficial if the animal is hungry, it may be harmful if sated. This assessment process is commonly referred to as the sense of valency (Ackley & Littman, 1991), which could be loosely defined as the internal process of that makes us perceive the effect of executing a

behavior as beneficial or harmful. The idea of a hormonally controlled learning method is inspired from this principle. The hormonal release is controlled by fluctuations of the agent's internal physiological variables, and only responds to positive compensations of any internal homeostatic variable and thus to contributions to physiological stability. Although we have not altered this basic rule, we suggest that it may be possible to modify both perception and behavior in an overall beneficial manner by evolving the valency criteria to facilitate adaptation in a larger variety of environments. First, the formulation of the hormonal response is intended to be an initial step toward this end. Also, this separation may facilitate the generalization of behavioral effects involving more than one homeostatic variable. Second, the learning process and therefore the assessment of the behavioral effect is framed by the notion of internal physiological stability formulated by Ashby (1965), formalized as an indicator of physiological stability. This notion establishes a basic condition for an agent to live, for example, an agent's overall behavioral pattern must perform to avoid a lethal boundary. Therefore, it establishes a quantifiable metric to assess the agent's performance. The hormonal response of our agent is also inspired by this notion, and provides a general context for the agent's perception and also for its behavior. Third, unlike the *drk* model, which disregards real perception, the learning architecture proposed here has a perception layer consisting of two concurrent processes: the adaptive topological network (GWRN) and the Hebbian learning of the affordance network. The unsupervised network can dynamically adapt its topology to the objects in the surrounding environment. If these changes are durable, their sensory perception will be captured by the GWRN and their related affordances will be learned by the hormonally controlled affordance learning. As the results demonstrate, if the representation is sufficiently accurate, the agent's perception adapts to the system, and provides grounded knowledge for the agent to maintain its physiological stability within the viable area.

Internal Physiology and Sensorimotor Interaction.

The results presented reflect the close link between the agent's internal physiology, characterized by the internal physiological decay and their environment, characterized by the distribution of resources. Although this does not intend to reject any other agent architec-

ture, it does reinforce the view of a single, ecological dynamics for an agent and its environment. In other words, this provides an implementation of the relationship between the resources afforded by the environment and the agent's internal rhythms. Importantly, this also relates to two major problems of robotics: building sensory representation and building a motivation-driven architecture. These two problems share the common ground of being very difficult when addressed separately. However, the results obtained from our experiments suggest that this difficulty may lessen when the knowledge learned is grounded by using a motivation-driven architecture, that is, when both problems are addressed simultaneously.

External Versus Internal Effect. This robotic environment has allowed us to investigate those behaviors and that part of the behavioral effect that can be perceived by changes in the agent's internal physiology as a method to learn object affordances of consummatory behaviors. Nevertheless, many, if not all, motor behaviors in the animal realm may also provoke an alteration of the physical environment, which may be perceived via visual or proprioceptive feedback. Thus, it seems reasonable that a possible extension of the architecture we presented here may include mechanisms such as the hormonal response to learn affordances by combining external and internal effect by using multiple sensory modalities. For example, the effect of shifting the position of an object reflects on changes in the optic flow, as it also correlates with our increase of fatigue as the task gradually extends over time. In a way, this future extension encompasses a potential manner to extend affordance learning from its simplest expression, maybe the *drk* model (Spier & McFarland, 1996), toward a multisensory model encompassing all sorts of behaviors.

Limitations. Most boundaries of this study are related to the divergence between the model and the animal models of adaptation. In this light, one of the main conceptual limitations is that the representation of the sensory is not directly based in optic flow, but on a series of snapshots taken from objects. We argue that although this procedure is not ideal, it has significantly reduced the complexity of the experimental set-up, while preserving the overall goal of the experiments, which is testing the adaptivity of the percep-

tion layers and the grounding principle. We would also like to stress that we do not claim a faithful biological compliance of the processes described, although the model has been strongly influenced by ethological and biological data. In the same manner, although the results relating behavior and physiological stability resulting from the experiments model physiological processes, they ought to be viewed in the context described and by the experimental set-up. A possible future extension of this work may consider the study of internal stability for the agent's physiological dynamics when the relationships between variables, drives, and behaviors are not univocal. Keeping in mind the goal of this article, this would not add any additional conceptual element to the procedure to learn grounded knowledge, but it would facilitate a complementary study on the internal physiology, which may in turn demand a more elaborated procedure to arbitrate the agent's behavior repertoire. Furthermore, some additional limitations may result from the use of a simulated robot. The perception and the interaction of a real robot with the real environment, even for the case of a relatively humble robot such as the Khepera, has a component of unreliability that simulators tend to overlook. The main elements of this study may be separated into two different groups: the homeostatic loop and sensorimotor interaction. The former depends on the intrinsic rate of consumption of the internal variables only, and would transfer to a real robot without any further changes. The latter may require some additional fine tuning because perception would be expected to be noisier and the interaction with the objects less reliable, although no conceptual changes would be required.

Behavior Selection. As mentioned in the introduction, this study does not address behavior selection per se. However, because we have selected a metric of performance based on physiological stability, it is necessary to endow the agent with a strategy to arbitrate among behaviors. Because the goal is to measure the performance provided by the perception mechanisms, we have chosen two very basic behavior-selection rules: motivation driven and affordance $\times$ drive (McFarland, 1993). The former disregards any knowledge from the environment, while the second utilizes the grounded knowledge learned about the surrounding environments.

7 Conclusion

This article has presented a method for learning object affordances based on the interaction between the sensorimotor and homeostatic loops. The learning principle is based on the internal reward generated by the effect of a consummatory behavior on the agent's homeostatic variables. The architecture proposed to learn the affordances consisted of the combination of a topological network, grouping together objects exhibiting similar perception patterns, and of an affordance network, connecting each node to every behavior. This second set of synapses, connecting to every behavior, encode the affordances offered to the agent by the objects represented by every topological node. The learning procedure consists of two concurrent processes. First, a sensory mapping process, which assesses the distance between the node closest and the sensory input. If the distance is larger than a certain threshold, the sensory pattern is considered underrepresented, and a novel node is added to the topological network. The synaptic weights connecting to the topological network are initialized to the average values of the two closest nodes and are fine tuned as new interactions with similar sensory patterns occur. Also, the nodes used less frequently are pruned from the topological map in the long run. Second, the reinforcing or weakening of the affordance synapses relating the topological nodes to the behaviors. This process is initiated by the hormonal release, which follows an interaction with an object in the environment if the object affords the behavior the agent executed. A successful interaction typically produces an increase of one or more of the agent's internal homeostatic variables, and consequently a hormonal release, which reinforces the synapse relating the active node to the behavior executed. Also, the nodes and synapses of the network, after the ecological principle, degrade if not used frequently.

We have tested this architecture in two different scenarios to investigate if the proposed sequence of processes relating sensorimotor interaction to the homeostatic loop can yield a neural substrate that learns affordances from objects in its environment. We have assessed both the statistical and behavioral performance of the proposed learning method. First, we have used statistical metrics to assess the accuracy required to represent the sensory patterns of each scenario by gradually increasing the network complexity

and assessing the resulting fitting error. Second, we have used the physiological stability as an ecological criterion to assess whether the learned affordances do significantly affect the agent's behavioral patterns and consequently its physiological stability. In particular, we tested the physiological variability indicator to assess the stability of the agent's internal resources under two behavior-selection policies, extracted from the ethological literature (McFarland, 1993). The results obtained suggest that the learning method effectively learns the behaviors afforded by every object and that the affordance $\times$ drive selection strategy produces behavioral patterns such that the agent's internal milieu is most stable. Therefore, the affordances learned from the environment can be used to arbitrate the agent's behavior repertoire in a way that maintains the internal physiological variables within the range of viability for that agent to survive. The learning method is relatively effective, since it can infer the affordances of new objects from the similarity to nearby nodes, representing similar sensory patterns. This greatly improves the efficiency of the learning algorithm, since the synaptic update, driven by a Hebbian process would require several interactions for fine-tuning.

This article presents a theoretical formalization of the dynamics between two interacting processes, governing developmental adaptation in the natural world: sensorimotor interaction and homeostasis. Furthermore, it presents the interaction between internal physiology and sensorimotor interaction as the foundation of grounded knowledge about certain elements of the environment. In other words, a foundation to learn the affordances in the context of artificial, embodied agents. We have investigated the nature of that interaction and the role internal motivations and external behavioral effect play in weakening or reinforcing the relationship between the agent's internal motivations and the environment. Therefore, this article is not addressing behavior selection or planning per se, but only perception as an agent's adaptive process to the environment—an adaptiveness that depends both on the change of the environment as well as on the internal assessment the agent makes of the behavioral effect (valency function/hormonal response). Affordances are learned from the relationship between the selection and execution of a behavior and the resulting physiological effect. We argue that the role of adaptive perception, as the capability of adapting the behavioral

patterns by the knowledge newly acquired from the environment through this method, provides an additional mechanism of adaptation to the environment which has not yet been properly harnessed.

As mentioned in the discussion, to build a representation of the world and to build a motivation-driven architecture separately is a difficult task. We believe that this article has provided the background and the experimental evidence to suggest that both aforementioned problems should be addressed simultaneously using a motivation-driven architecture. If there is some consensus on the fact that these processes occur simultaneously in the animal realm and that value is learned on the grounds of animal urges for action, this may also suggest that an agent design that considers both internal motivations and affordance learning simultaneously, may also benefit from the same inspiration then those using rigid schemas viewing perception and action as sequential processes.

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References

- Ackley, D., & Littman, M. (1991). Interactions between learning and evolution. In C. Langton, C. Taylor, J. D. Farmer, & S. Rasmussen (Eds.), *Artificial life II, SFI studies in the sciences of complexity* (Vol. X, pp. 487–509). Menlo Park, CA: Addison-Wesley.
- Ashby, W. (1965). *Design for a brain: The origin of adaptive behaviour*. London: Chapman & Hall.
- Avila-García, O., & Cañamero, L. (2002). A comparison of behaviour selection architectures using viability indicators. In *Proceedings of International Workshop on Biologically Inspired Robotics: The Legacy of W. Grey Walter*. Bristol HP Labs, UK.
- Baecker, R., Grudin, J., Buxton, W. A., & Greenberg, S. (1995). *Readings in human-computer interaction: Towards the year 2000*. San Francisco, CA: Morgan Kaufmann.
- Bindra, D. (1969). The interrelated mechanisms of reinforcement and motivation, and the nature of their influence on response. In W. J. Arnold & D. Levine (Eds.), *Nebraska symposium on motivation* (pp. 1–33). University of Nebraska Press.
- Cannon, W. B. (1929). *The wisdom of the body*. London: Kegan Paul, Trench, Trubner & Co.
- Chemero, A., & Turvey, M. T. (2007). Gibsonian affordances for roboticists. *Adaptive Behavior*, 15(4), 473–480.
- Cisek, P. (2007). Cortical mechanisms of action selection: The affordance competition hypothesis. *Philosophical Transactions of the Royal Society B*, 362, 1585–1599.
- Cisek, P., & Kalaska, J. (2005). Neural correlates of reaching decisions in dorsal premotor cortex: specification of multiple direction choices and final selection of action. *Neuron*, 45(5), 801–814.
- Cooper, R., & Glasspool, D. (2002). Learning affordances and action schemas. In R. French & J. Sougne (Eds.), *Connectionist models of learning, development and evolution* (pp. 133–142). London: Springer-Verlag.
- Cos-Aguilera, I., Cañamero, L., & Hayes, G. (2003a). Motivation-driven learning of object affordances: First experiments using a simulated khepera robot. In F. Detje, D. Dörner, & H. Schaub (Eds.), *The logic of cognitive systems. Proceedings of the 5th International Conference on Cognitive Modelling, ICCM* (pp. 57–62). Universitäts-Verlag Bamberg.
- Cos-Aguilera, I., Cañamero, L., & Hayes, G. M. (2003b). Learning object functionalities in the context of behavior selection. In U. Nehmzow & C. Melhuish (Eds.), in *Proceedings of the 3rd Conference Towards Intelligent Mobile Robotics*. (pp. 9–14)
- Cos-Aguilera, I., Cañamero, L., & Hayes, G. M. (2004). Using a sofim to learn object affordances. In J. L. De la Rosa & J. A. Ramón-Pagés (Eds.), in *Proceedings of the 5th Workshop of Physical Agents (WAF04)*. (pp. 139–150). University of Girona.
- Damoulas, T., Cos-Aguilera, I., Hayes, G., & Taylor, T. (2005). Valency for adaptive homeostatic agents: Relating evolution and learning. In *Proceedings of the 8th European Conference on Artificial Life (ECAL2005)*, Lecture Notes in Artificial Intelligence (Vol. 3630, pp. 936–945). Berlin/Heidelberg: Springer.
- Doğar, M., Ugur, E., Şahin, E., & Çakmak, M. (2008). Using learned affordances for robotic behavior development. In *Proceedings of the 2008 IEEE International Conference on Robotics and Automation*. (pp. 3802–3807). Pasadena, CA: IEEE
- Duchon, A. P., & Warren, W. H. (1994). Robot navigation from a Gibsonian viewpoint. In *Proceedings of IEEE International Conference on Systems, Man and Cybernetics* (pp. 2–5). IEEE.
- Duchon, A. P., Warren, W. H., & Kaelbling, L. P. (1995). Ecological robotics: Controlling behaviour with optical flow. In J. Moore & J. Lehman (Eds.), *Proceedings of the 17th*

- Annual Cognitive Science Conference*. (pp. 164–169). Mahwah, NJ: Lawrence Erlbaum Associates.
- Fagg, A. H., & Arbib, M. A. (1998). Modeling parietal-premotor interactions in primate control of grasping. *Neural Networks*, 11(7–8), 1277–1303.
- Fellous, J.-M. (2001). Dopamine modulation of prefrontal delay activity-reverberatory activity and sharpness of tuning curves. *Neurocomputing*, 38–40, 1549–1556.
- Fellous, J.-M. (2004). From human emotions to robotic emotions. In L. Cañamero & E. Hudlika (Eds.), *Architectures for modelling emotion: Cross-disciplinary foundations. Papers from the 2004 AAAI Spring Symposium*. Menlo Park, CA: AAAI Press.
- Fellous, J.-M., & Suri, R. E. (2003). *The handbook of brain theory and neural networks* (chap. The Roles of Dopamine). Cambridge, MA: MIT Press.
- Freud, S. (1940). *An outline of psychoanalysis*. London: Hogarth Press.
- Gibson, J. J. (1986). *The ecological approach to visual perception*. Hillsdale, N.J./London.
- Harnad, S. (1990). The symbol grounding problem. *Physica D*, 42, 335–346.
- Hebb, D. (1949). *The organization of behavior. A neuropsychological theory*. New York: Wiley.
- Hinde, R. A. (1960). Energy models of motivation. In *Symposium of Society of Experimental Biology* (Vol. 14, pp. 199–213). Cambridge University Press.
- Houk, J. C., Adams, J. L., & Barto, A. G. (1995). A model of how the basal ganglia generate and use neural signals that predict reinforcement. In J. C. Houk, J. L. Davis, & D. G. Beiser (Eds.), *Models of information processing in the basal ganglia* (2nd ed., 1998; pp. 249–270). Cambridge, MA: MIT Press/Bradford Books.
- Hull, C. (1943). *Principles of behaviour: an introduction to behaviour theory*. New York, NY: D. Appleton-Century Company, Inc.
- Khamassi, M., Lachèze, L., Girard, B., & Guillot, A. (2006). Actor-critic models of reinforcement learning in the basal ganglia: From natural to artificial rats. *Adaptive Behavior*, 13(2), 131–148.
- Konidaris, G., & Barto, A. (2006). An adaptive robot motivational system. In S. Nolfi, G. Baldassarre, R. Calabretta, J. C. T. Hallam, D. Marocco, J.-A. Meyer, O. Miglino, & D. Parisi (Eds.), *Animals to animats 9: Proceedings of the 9th International Conference on Simulation of Adaptive Behavior* (pp. 346–356). Berlin: Springer Verlag.
- Konidaris, G., & Hayes, G. M. (2005). An architecture for behavior-based reinforcement learning. *Adaptive Behavior*, 69, 861–872.
- Kriekhaus, E., & Wolf, G. (1968). Acquisition sodium by rats: Interaction of innate mechanisms and latent learning. *Journal of comparative and Physiological Psychology*, 65, 197–201.
- Lorenz, K. (1966). *Evolution and modification of behaviour*. London: Methuen & Co Ltd.
- Marrow, A. (1969). *The practical theorist: The life and work of Kurt Levin*. New York: Basic Books.
- Marsland, S., Shapiro, J., & Nehmzow, U. (2002). A self-organising neural network that grows when required. *Neural Networks*, 15(8–9), 1041–1058.
- Matarić, M., & Brooks, R. (1990). *Cambrian intelligence: The early history of the new AI* (chap. Learning a distributed map representation based on navigation behaviors). Cambridge, MA: MIT Press.
- Maturana, H. R., & Varela, F. J. (1980). *Boston Studies in the Philosophy of Science: Vol. 42. Autopoiesis and cognition: The realization of the living*. Dordrecht/London: Reidel.
- McDougall, W. (1913). The sources and direction of psychophysical energy. *American Journal of Insanity*.
- McFarland, D. (1993). *Animal behaviour* (3rd ed.). Harlow: Prentice Hall.
- McFarland, D. J., & Sibly, R. M. (1975). The behavioural final common path. *Proceedings of the Royal Society of London*, 270, 265–293.
- Morineau, T., & Frénod, E. (2009). Affordance modelling and simulation in virtual environment. *European Journal of Scientific Research*, 33(1), 130–143.
- Nehaniv, C., & Dautenhahn, K. (1998). Mapping between dissimilar bodies: Affordances and the algebraic foundations of imitation. In J. Demiris & A. Birk (Eds.), *Proceedings of the European Workshop on Learning Robots 1998 (EWLR-7)* (pp. 177–184) Edinburgh, UK.
- Redgrave, P., Prescott, T., & Gurney, K. (1999). The basal ganglia: a vertebrate solution to the selection problem? *Neuroscience*, 89, 1009–1023.
- Şahin, E., Çakmak, M., Doğar, M. R., Ugur, E., & Üçoluk, G. (2007). To afford or not to afford: A new formalization of affordances toward affordance-based robot control. *Adaptive Behavior*, 15, 447–472.
- Song, J., & Nakayama, K. (2007). Automatic adjustment of visuomotor readiness. *Journal of Vision*, 7(5), 1–9.
- Spier, E., & McFarland, D. (1996). A finer-grained motivational model of behaviour sequencing. In P. Maes, M. J. Mataric, J.-A. Meyer, J. Pollack, & S. W. Wilson (Eds.), *A Finer-Grained Motivational Model of Behaviour Sequencing. From Animals to Animats 4: Proceedings of SAB96* (pp. 255–263). Cambridge, MA: MIT Press.
- Spier, E., & McFarland, D. (1997). Possibly optimal decision-making under self-sufficiency and autonomy. *Journal of Theoretical Biology*, 189, 317–331.
- St. Amant, R. (1999). User interface affordances in a planning representation. *Human-Computer Interaction*, 14(3), 317–354.

- Stoffregen, T. (2003). Affordances as properties of the animal environment system. *Ecological Psychology*, 15, 115–134.
- Sutton, R., & Barto, A. (1981). Towards a modern theory of adaptive networks: Expectation and prediction. *Psychological Review*, 198(88), 135–170.
- Tinbergen, N. (1951). *The study of instinct*. Oxford: Oxford University Press.
- Turvey, M. (1992). Affordances and prospective control: an outline of the ontology. *Ecological Psychology*, 4, 271–281.
- Usher, M., & Davelaar, E. J. (2002). Neuromodulation of decision and response selection. *Neural Networks*, 15, 635–645.
- Widrow, B., & Hoff, M. (1960). Adaptive switching circuits. In *1960 IRE WESCON Convention Record (pt. 4)* (pp. 96–104). IRE

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